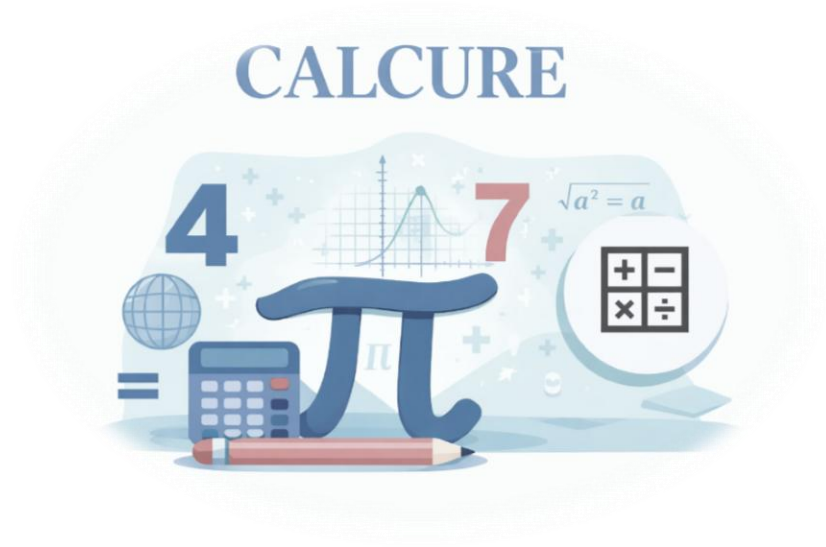

Software Requirements Specification for



Submitted to

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Table of Contents

1.Introduction.....	4
1.1 Purpose.....	4
1.2 Project Scope.....	4
1.3 Glossary	5
1.4 References	6
1.5 Overview	6
2.0 Stakeholders	6
2.1 Primary Stakeholders	6
2.1.1 Student	6
2.1.2 Teacher	7
2.1.3 Moderator.....	7
2.1.4 Administrator	8
2.1.5 Job Seeker	8
2.2 Secondary / External Stakeholders	8
2.2.1 Guardian.....	8
2.2.2 Educational Partner Platform	9
2.2.3 Project Owner (Authority)	9
3. Requirement Specification.....	9
3.1 Business Requirements	9
3.2 Business Objectives	10
3.3 User Requirements.....	11
3.3.1 Primary Users.....	11
3.3.2 Secondary Users.....	12
3.4 Business Rules	13
3.5 Functional Requirements	14
3.6 Quality Attributes / Performance Requirements	16
3.7 External Interface Requirements.....	17
3.8 Design Constraints	18
3.9 Data Requirements.....	19
4. Requirement Engineering Process	20
4.1 Requirement Elicitation Techniques	20
4.1.1 Interviews.....	20
4.1.2 Workshops.....	21
4.1.3 Focus Group Discussion	22
4.1.4 Observation	22

4.1.5 Questionnaires.....	23
4.1.6 User Interface Analysis	24
4.2 Sample of Requirement Collection	24
4.3 Requirement Validation	25
4.3.1 Review the Requirements	25
4.3.2 Test the Requirements	25
4.3.3 Simulate the Requirements	25
5. Use Case Diagram.....	26
6. Use case Description	27
6.1 UC -01 : Authentication	27
6.2 UC -02 : Authorization	28
6.3 UC -03 : View Tutorial	29
6.4 UC -04 : Attempt Quiz.....	30
6.5 UC -05 : Submit Assignment.....	31
6.6 UC -06 : Practice Exercise	33
6.7 UC -07 : Track Progress.....	34
6.8 UC -08 : Manage Subscription.....	35
6.9 UC -09 : Upload Content	37
6.10 UC -10 : Manage Notes	38
6.11 UC -11 : Assign Task.....	39
6.12 UC -12 : Notify Guardian	40
6.13 UC -13 : Generate Performance Report.....	41
6.14 UC -14 : Receive Performance Report	43
6.15 UC -15 : Manage Institutional Account.....	44
6.16 UC -16 : Browse Free Content.....	45
6.17 UC -17 : Register User.....	46
6.18 UC -18 : Manage Member	47
6.19 UC -19 : Review Content.....	48
6.20 UC -20 : Approve/ Reject Content.....	49
6.21 UC -21 : Process Payment	50
6.22 UC -22 : Send Notification	51
7. Activity Diagrams	53
8. DFD.....	75
9. RTM (Requirement Traceability Matrix):.....	76

Figure of Contents

Figure 1: Use Case Diagram For Calcure	26
Figure 2: Activity diagram for Authentication	53
Figure 3: Activity diagram for Authorization	54
Figure 4: Activity diagram for View Tutorial	55
Figure 5: Activity Diagram for Attempt Quiz	56
Figure 6: Activity Diagram for Submit Assignment	57
Figure 7: Activity Diagram for Practice Exercise	58
Figure 8: Activity Diagram for Track Progress.....	59
Figure 9: Activity Diagram for Manage Subscription	60
Figure 10: Activity Diagram for Upload Content	61
Figure 11: Activity Diagram for Manage Notes.....	62
Figure 12: Activity Diagram for Assign Task	63
Figure 13: Activity Diagram for Notify Guardian	64
Figure 14 : Activity Diagram for Generate Performance Report.....	65
Figure 15 : Activity Diagram for Receive Performance Report	66
Figure 16: Activity Diagram for Manage Institutional Account	67
Figure 17: Activity Diagram for Browse Free Content.....	68
Figure 18: Activity Diagram for Register User.....	69
Figure 19: Activity Diagram for Manage Member	70
Figure 20: Activity Diagram for Review Content.....	71
Figure 21: Activity Diagram for Approve/Reject Content.....	72
Figure 22: Activity Diagram for Process Payment	73
Figure 23: Activity Diagram for Sent Notification	74
Figure 24: Level 0 DFD	75

1.Introduction

Calcure is an online mathematics learning platform designed to support students in understanding, practicing, and mastering mathematical concepts through **interactive and guided learning tools**. The system aims to simplify complex mathematical problems by providing **step-by-step solutions, visual representations, and video tutorials** that help learners grasp concepts more effectively.

By integrating **dynamic visualizations** such as graphs, diagrams, and animations, Calcure enhances conceptual clarity and promotes logical reasoning. The inclusion of **video-based explanations** allows students to learn at their own pace, revisit difficult topics, and strengthen foundational knowledge. Instant feedback and performance tracking features further assist students in identifying their strengths and areas for improvement.

Overall, Calcure seeks to reduce students' fear of mathematics and encourage analytical thinking by offering an **engaging, user-friendly, and self-paced educational environment** that supports diverse learning styles.

1.1 Purpose

The purpose of this document is to provide a detailed and structured description of the **Calcure online mathematical educational system**. It outlines the system's objectives, core features, functional requirements, and constraints to ensure a clear understanding among all stakeholders, including students, teachers, administrators, moderators, and the project authority.

This document serves as a reference for system design and development by defining how Calcure will support learning through **step-by-step problem-solving, visualizations, and video tutorials**. It also acts as a formal agreement between the project owner and the development team by specifying the expected behavior and performance of the system.

The intended audience of this Software Requirements Specification (SRS) includes the **project team, project supervisor, client, and other relevant stakeholders** involved in the development, evaluation, and approval of the system.

1.2 Project Scope

Calcure is an online, interactive mathematics learning platform designed to assist students in learning and practicing mathematical concepts in a structured, engaging, and self-paced manner. The system focuses on reducing conceptual difficulty and learning anxiety by breaking down problems into clear step-by-step solutions supported by visual explanations and video-based tutorials.

The platform enables students to:

- ❖ Learn mathematical concepts through guided steps and visualizations
- ❖ Watch video tutorials for detailed topic explanations
- ❖ Practice problems and receive instant feedback
- ❖ Track learning progress and performance over time

Teachers and content creators can upload and manage:

- ❖ Step-by-step solution content
- ❖ Visual learning materials (graphs, diagrams, animations)
- ❖ Video tutorials aligned with the curriculum

Moderators and administrators ensure content quality, system security, and smooth platform operation. Guardians and project authorities can monitor learning outcomes and system performance through reports and analytics.

Overall, Calcure aims to create a comprehensive digital learning ecosystem that supports multiple learning styles, enhances conceptual understanding, and promotes logical and analytical thinking in mathematics.

1.3 Glossary

Term	Definition
Calcure	The mathematical educational software system
User	A student who uses the system to practice mathematics
Problem	A mathematical question presented to the user
Solution	Step-by-step explanation of a mathematical problem
Module	A specific mathematics topic (e.g., Algebra, Arithmetic)
Attempt	A single submission of an answer by the user
Score	Numerical value representing performance
Feedback	System response indicating correctness of an answer
Dashboard	Main interface displaying progress and results
Progress Tracking	Monitoring improvement over time
Difficulty Level	Classification of problems (Easy, Medium, Hard)
Timer	Time limit set for solving problems
Admin	Authorized person managing system content
Database	Collection of stored users, problems, and results

Software Requirements Specification (SRS)	Document describing system functionality and constraints
Stakeholder	Any person with interest in the Calcure project

1.4 References

Standards and Textbooks:

- ❖ IEEE Std 830-1998 – Software Requirements Specification
- ❖ ISO/IEC/IEEE 29148:2018 – Requirements Engineering
- ❖ Ian Sommerville, Software Engineering, Pearson
- ❖ Discrete Mathematics and Its Applications – Rosen

Educational Sources:

- ❖ National Curriculum and Textbook Board (NCTB), Bangladesh
- ❖ Khan Academy (Concept Reference)
- ❖ NCERT Mathematics Framework

1.5 Overview

Calcure is an integrated mathematical learning platform designed to bridge the gap between traditional education and modern digital learning. By combining structured content, verified teaching materials, interactive assessments, and performance analytics, Calcure aims to create a reliable learning environment for students, teachers, and institutions alike.

2.0 Stakeholders

The Calcure system involves multiple stakeholders, each with distinct roles, responsibilities, and expectations. Identifying these stakeholders is essential to ensure that the system is designed in a **user-centered, efficient, and sustainable manner**, while meeting educational, technical, and administrative requirements.

2.1 Primary Stakeholders

2.1.1 Student

Description:

Students are the primary end-users of Calcure. They use the system to learn mathematics through step-by-step solutions, visual explanations, interactive exercises, and video tutorials.

Objectives / Needs:

- ❖ Access structured lessons, practice problems, and guided solutions
- ❖ Learn using visualizations and video tutorials
- ❖ Receive instant feedback on answers
- ❖ Track academic progress and performance
- ❖ Use the system in both Bangla and English

Responsibilities:

- ❖ Participate in learning activities honestly
- ❖ Maintain accurate personal information
- ❖ Follow platform rules and usage guidelines

2.1.2 Teacher

Description:

Teachers are subject matter experts responsible for creating, managing, and updating mathematical content within the system.

Objectives / Needs:

- ❖ Upload exercises, step-by-step solutions, and visual learning materials
- ❖ Provide video tutorials and explanations
- ❖ Monitor student performance and learning trends

Responsibilities:

- ❖ Ensure accuracy and clarity of content
- ❖ Provide timely guidance and feedback
- ❖ Maintain professional and ethical standards

2.1.3 Moderator

Description:

Moderators ensure the quality and reliability of the platform by reviewing content and addressing user-reported issues.

Objectives / Needs:

- ❖ Approve or reject educational content
- ❖ Maintain consistency and quality across materials
- ❖ Resolve disputes or flagged activities

Responsibilities:

- ❖ Review submitted exercises and solutions
- ❖ Enforce platform policies
- ❖ Handle reported issues efficiently

2.1.4 Administrator

Description:

Administrators manage the overall operation, security, and configuration of the Calcure system.

Objectives / Needs:

- ❖ Maintain system availability and performance
- ❖ Manage users, roles, and permissions
- ❖ Ensure data security and system integrity

Responsibilities:

- ❖ Monitor system operations and logs
- ❖ Perform backups and updates
- ❖ Resolve technical and access-related issues

2.1.5 Job Seeker

Description:

Individuals seeking teaching, content creation, or moderation roles within the Calcure platform.

Objectives / Needs:

- ❖ Apply for available roles
- ❖ Submit qualifications and credentials
- ❖ Track application status

Responsibilities:

- ❖ Provide accurate and verifiable information
- ❖ Communicate professionally through the platform

2.2 Secondary / External Stakeholders

2.2.1 Guardian

Description:

Parents or guardians who monitor students’ academic progress and engagement.

Objectives / Needs:

- ❖ View student performance reports
- ❖ Receive progress updates and alerts

Responsibilities:

- ❖ Support and motivate students
- ❖ Encourage consistent and responsible learning

2.2.2 Educational Partner Platform

Description:

External educational organizations or platforms that may provide supplementary learning resources or content validation.

Objectives / Needs:

- ❖ Share educational materials
- ❖ Integrate compatible content

Responsibilities:

- ❖ Ensure data privacy and security
- ❖ Maintain academic accuracy and relevance

2.2.3 Project Owner (Authority)

Description:

The authority or organization responsible for overseeing the Calcure project and approving strategic decisions.

Objectives / Needs:

- ❖ Monitor system performance and impact
- ❖ Approve major updates and policies
- ❖ Ensure alignment with educational goals

Responsibilities:

- ❖ Provide strategic direction
- ❖ Approve system enhancements and expansions

3. Requirement Specification

All the requirements based on the elicitation process are described in this section.

3.1 Business Requirements

BR-1: Provide subscription-based math learning services

Field	Content	Field	Content
BR-1	The system shall provide subscription-based math learning services for individual learners and educational institutions.		
Description	The system shall offer free and premium access levels for learning resources across Arithmetic, Algebra, Geometry,		

	Trigonometry, Calculus, and Statistics to support sustainable revenue.		
Stakeholders	Owner, Admin, Student, Institution	Priority	High

BR-2: Enable verified content marketplace with partners and teachers

Field	Content	Field	Content
BR-2	The system shall enable teachers and partner platforms to publish verified learning content under owner control.		
Description	The system shall support submission, review, approval, publishing, and performance tracking to allow remote teaching and partner integration.		
Stakeholders	Owner, Admin, Teacher, Partner, Moderator	Priority	High

3.2 Business Objectives**BO-1: Improve math learning outcomes through guided practice**

Field	Content	Field	Content
BO-1	The system shall improve learners' mathematics performance through structured lessons, step-by-step solutions, and continuous assessment.		
Description	Calcure shall support topic-wise learning with quizzes/practice and progress tracking so learners can identify weak areas and improve over time.		
Stakeholders	Student, Job Seeker, Teacher, Owner	Priority	High

BO-2: Expand reach of quality math education via remote delivery and partnerships

Field	Content	Field	Content
BO-2	The system shall enable scalable delivery of verified math content to schools/colleges and individuals through subscriptions and partner onboarding.		
Description	Calcure shall support institution subscriptions, remote teaching, and moderated content publishing to ensure quality while growing user base and revenue.		
Stakeholders	Owner, Admin, Institution, Teacher, Partner	Priority	High

3.3 User Requirements

3.3.1 Primary Users

UR-P1: Student subscription access to premium resources

Field	Content	Field	Content
UR-P1	The system shall allow students to access premium learning resources based on an active subscription or institutional entitlement.		
Description	Premium resources include video tutorials, step-by-step solutions, notes, quizzes, practice sets, and assignments.		
Stakeholders	Student, Admin, Owner	Priority	High

UR-P2: Topic-wise learning navigation

Field	Content	Field	Content
UR-P2	The system shall allow students to browse learning content by education level, subject, and topic.		
Description	Students shall be able to locate lessons for relevant classes/levels and view connected resources per topic.		
Stakeholders	Student, Teacher	Priority	High

UR-P3: Step-by-step solution consumption

Field	Content	Field	Content
UR-P3	The system shall allow students to view step-by-step solutions for textbook and practice problems.		
Description	Solutions shall be presented as ordered steps with mathematical notation, explanation, and final answer.		
Stakeholders	Student, Teacher	Priority	High

UR-P4: Video tutorial learning

Field	Content	Field	Content
UR-P4	The system shall allow students to watch topic-based video tutorials with related notes and materials.		
Description	Students shall be able to view approved lesson videos and access attached notes/solutions for the same topic.		
Stakeholders	Student, Teacher	Priority	High

UR-P5: Assessment and practice participation

Field	Content	Field	Content
UR-P5	The system shall allow students to attempt quizzes and practice sets and receive performance results.		
Description	The system shall store attempts and show scores and feedback according to quiz rules set by the teacher.		
Stakeholders	Student, Teacher	Priority	High

UR-P6: Teacher content and assessment creation

Field	Content	Field	Content
UR-P6	The system shall allow teachers to upload tutorials, notes, step solutions, and create quizzes/assignments for topics.		
Description	Teachers shall be able to submit materials for review and assign practice to learners after completing a topic.		
Stakeholders	Teacher, Student, Moderator	Priority	High

3.3.2 Secondary Users**UR-S1: Guest access to free features**

Field	Content	Field	Content
UR-S1	The system shall allow guest users to browse and use selected free features without account creation.		
Description	Guests shall be able to preview limited lessons/resources while premium content remains restricted.		
Stakeholders	Guest, Owner	Priority	Medium

UR-S2: Guardian progress reports

Field	Content	Field	Content
UR-S2	The system shall allow guardians to receive student performance summaries through SMS and/or email.		
Description	Reports shall include progress, quiz outcomes, and assignment status as configured by teachers/admins.		
Stakeholders	Guardian, Student, Teacher, Admin	Priority	Medium

UR-S3: Institution subscription and student access control

Field	Content	Field	Content
UR-S3	The system shall allow institutions to subscribe and provide entitled access to enrolled learners.		
Description	Institution representatives shall be able to manage rosters/classes and apply institutional access rights.		
Stakeholders	Institution Admin, Admin, Owner	Priority	High

UR-S4: Job seeker preparation support

Field	Content	Field	Content
UR-S4	The system shall allow job seekers to use learning content and practice assessments for preparation.		
Description	The system shall provide topic-wise practice and progress tracking suitable for competitive exam preparation.		
Stakeholders	Job Seeker, Owner	Priority	Medium

3.4 Business Rules**BRule-1: Only approved content may be published**

Field	Content	Field	Content
BRule-1	The system shall publish learning materials only after moderator approval.		
Description	Submitted videos/notes/solutions/quizzes shall remain unavailable to learners until approved; rejected items shall require resubmission after correction.		
Stakeholders	Moderator, Teacher, Owner	Priority	High

BRule-2: Premium access requires valid entitlement

Field	Content	Field	Content
BRule-2	The system shall restrict premium resources to users with an active subscription or institutional entitlement.		
Description	The system shall enforce entitlement checks for videos, quizzes, practice sets, notes, assignments, and step-by-step solutions.		
Stakeholders	Student, Admin, Owner	Priority	High

3.5 Functional Requirements

FR-1: Account creation and role-based authorization

Field	Content	Field	Content
FR-1	The system shall support account creation and role-based authorization for Student, Teacher, Moderator, Admin, and Owner roles.		
Description	The system shall enforce permissions for content creation, moderation, membership management, and learning access based on assigned roles.		
Stakeholders	All Users, Admin	Priority	High

FR-2: Admin membership management

Field	Content	Field	Content
FR-2	The system shall allow admins to create, update, suspend, and assign roles to members.		
Description	Member profiles shall include phone number, email address, educational qualifications (for teachers/moderators), and relevant attributes.		
Stakeholders	Admin, Owner	Priority	High

FR-3: Teacher lesson and content upload

Field	Content	Field	Content
FR-3	The system shall allow teachers to create lessons and upload videos, notes, and step-by-step solutions with required metadata.		
Description	Metadata shall include subject, topic, education level, tags, and content type to support browsing and review.		
Stakeholders	Teacher, Admin	Priority	High

FR-4: Moderation workflow management

Field	Content	Field	Content
FR-4	The system shall provide a moderation workflow for reviewing, approving, rejecting, and publishing submitted materials.		
Description	Moderators shall record decisions with feedback; the system shall maintain status history for each submitted item.		
Stakeholders	Moderator, Teacher, Owner	Priority	High

FR-5: Content catalog browsing and search

Field	Content	Field	Content
FR-5	The system shall provide browsing and search for lessons and resources by subject, topic, level, and content type.		
Description	The system shall support filters and sorting to quickly find relevant tutorials, notes, and solutions.		
Stakeholders	Student, Guest, Job Seeker	Priority	Medium

FR-6: Video delivery and playback support

Field	Content	Field	Content
FR-6	The system shall deliver approved video tutorials with adaptive streaming support.		
Description	The system shall support multiple quality levels and stable playback suitable for varying network conditions.		
Stakeholders	Student, Teacher	Priority	High

FR-7: Step-by-step solution rendering

Field	Content	Field	Content
FR-7	The system shall display step-by-step solutions using consistent mathematical notation and formatting.		
Description	The system shall support equations/symbols and structured steps for clarity and correctness.		
Stakeholders	Student, Teacher	Priority	High

FR-8: Quiz management and attempt processing

Field	Content	Field	Content
FR-8	The system shall allow teachers to create quizzes and allow learners to attempt them with automatic scoring where applicable.		
Description	The system shall store attempt details, scores, and allow configured feedback/solution visibility rules.		
Stakeholders	Teacher, Student	Priority	High

FR-9: Assignments and submissions

Field	Content	Field	Content
FR-9	The system shall allow teachers to publish assignments/practice sets and allow students to submit responses before deadlines.		

Description	The system shall track submission status, timestamps, and teacher feedback/grades (manual or automated where applicable).		
Stakeholders	Teacher, Student	Priority	High

FR-10: Progress tracking and reporting

Field	Content	Field	Content
FR-10	The system shall track learner progress and generate performance reports for students, teachers, and guardians.		
Description	The system shall compute progress metrics from activity (content completion, quiz outcomes, assignment status) and support scheduled delivery to guardians via configured channels.		
Stakeholders	Student, Teacher, Guardian, Admin	Priority	High

3.6 Quality Attributes / Performance Requirements**QA-1: Availability**

Field	Content	Field	Content
QA-1	The system shall be available for end users with high uptime.		
Description	Target availability shall be at least 99.5% monthly for core learning and subscription functions.		
Stakeholders	All Users, Owner	Priority	High

QA-2: Response time

Field	Content	Field	Content
QA-2	The system shall provide acceptable response time for typical user actions.		
Description	For standard operations (browse, open lesson, submit quiz answer), the system should respond within 2 seconds under normal load.		
Stakeholders	Student, Teacher	Priority	High

QA-3: Streaming performance (low bandwidth support)

Field	Content	Field	Content
QA-3	The system shall support video streaming performance suitable for low and variable bandwidth.		
Description	The system shall provide adaptive quality and minimize buffering for typical mobile networks.		
Stakeholders	Student, Job Seeker	Priority	High

QA-4: Security and privacy

Field	Content	Field	Content
QA-4	The system shall protect user data and prevent unauthorized access to content and personal information.		
Description	The system shall enforce role-based permissions, secure data transmission, and maintain auditable administrative actions.		
Stakeholders	Admin, Owner, All Users	Priority	High

QA-5: Scalability

Field	Content	Field	Content
QA-5	The system shall scale to support growth in users, content volume, and concurrent quiz attempts.		
Description	The system shall maintain stable performance during peak usage through scalable deployment and caching where applicable.		
Stakeholders	Owner, Admin	Priority	Medium

3.7 External Interface Requirements**EIR-1: Payment gateway interface**

Field	Content	Field	Content
EIR-1	The system shall interface with external payment gateways to process subscription fees.		
Description	The system shall support payment confirmation callbacks/webhooks and store transaction references for audit.		
Stakeholders	Student, Admin, Owner	Priority	High

EIR-2: SMS/Email service interface

Field	Content	Field	Content
EIR-2	The system shall interface with SMS and email service providers to deliver notifications and guardian reports.		
Description	The system shall support templates, delivery status tracking, and retry handling for failed deliveries.		
Stakeholders	Guardian, Teacher, Admin	Priority	Medium

3.8 Design Constraints

DC-1: Mobile-first user experience

Field	Content	Field	Content
DC-1	The system shall be designed with a mobile-first user interface.		
Description	The UI shall remain usable on common Android devices and smaller screens.		
Stakeholders	Student, Owner	Priority	High

DC-2: Bilingual support

Field	Content	Field	Content
DC-2	The system shall support Bangla and English for the user interface and content metadata.		
Description	The system shall allow bilingual titles/descriptions and support future localization expansion.		
Stakeholders	Student, Teacher	Priority	Medium

DC-3: Standard math notation rendering

Field	Content	Field	Content
DC-3	The system shall render mathematical expressions using a standard typesetting method.		
Description	Mathematical symbols and equations shall display consistently across supported devices and platforms.		
Stakeholders	Student, Teacher	Priority	High

DC-4: Content versioning and traceability

Field	Content	Field	Content
DC-4	The system shall maintain version history for learning materials and review decisions.		
Description	The system shall retain who submitted/approved/updated content and when, to support accountability and rollback.		
Stakeholders	Moderator, Admin, Owner	Priority	High

DC-5: Protection for minors' data

Field	Content	Field	Content
DC-5	The system shall minimize and protect personal data for minor students.		

Description	The system shall restrict exposure of student personal information and support guardian-linked reporting.		
Stakeholders	Guardian, Admin, Owner	Priority	High

3.9 Data Requirements

DR-1: User and role data

Field	Content	Field	Content
DR-1	The system shall store user profile data and role assignments.		
Description	Data shall include phone number, email address, role, and teacher qualifications where applicable.		
Stakeholders	Admin, Owner	Priority	High

DR-2: Content metadata and storage references

Field	Content	Field	Content
DR-2	The system shall store content metadata and file/storage references for videos, notes, and solutions.		
Description	Metadata shall include subject, topic, level, author, timestamps, and publication status.		
Stakeholders	Teacher, Moderator	Priority	High

DR-3: Assessment and submission records

Field	Content	Field	Content
DR-3	The system shall store quiz attempts, scores, answers, and assignment submissions.		
Description	Records shall support progress analytics, reporting, and review of past performance.		
Stakeholders	Student, Teacher, Guardian	Priority	High

DR-4: Subscription and payment records

Field	Content	Field	Content
DR-4	The system shall store subscription status, entitlement periods, and payment transaction references.		
Description	Data shall support access control, billing history, refunds (if applicable), and audit requirements.		
Stakeholders	Student, Admin, Owner	Priority	High

DR-5: Notification and delivery history

Field	Content	Field	Content
DR-5	The system shall store notification content references and delivery status for SMS/email.		
Description	The system shall retain recipient, timestamp, channel, provider response, and retry outcomes for reporting.		
Stakeholders	Guardian, Admin	Priority	Medium

4. Requirement Engineering Process

The Requirement Engineering (RE) process for the **Calcure** project follows a structured and iterative approach involving requirement elicitation, analysis, validation, and documentation. The goal of this process is to ensure that Calcure effectively meets the educational needs of students while also supporting teachers and administrators through well-defined and reliable system functionalities.

The Calcure development team adopted an **incremental requirement engineering model**, emphasizing continuous stakeholder involvement, feedback through prototypes, and traceable refinement of requirements. This approach ensures close alignment between academic goals, usability expectations, and technical feasibility throughout the system development lifecycle.

4.1 Requirement Elicitation Techniques

Requirement elicitation for Calcure employed multiple complementary techniques to ensure accurate understanding, completeness, and stakeholder consensus. Each technique targeted specific stakeholder groups such as students, teachers, content creators, and administrators.

4.1.1 Interviews**Purpose:**

To collect in-depth qualitative insights regarding difficulties in learning mathematics, preferred learning methods, and expectations from an online learning platform.

Approach:

- ❖ Semi-structured one-on-one interviews
- ❖ Conducted both face-to-face and online
- ❖ Duration: 20–30 minutes per participant

Stakeholders Involved:

- ❖ 12 students (primary and secondary levels)
- ❖ 2 mathematics teachers

Example Interview Questions:

1. Which math topics do you find the most difficult?
2. Do you feel comfortable learning math by watching videos?
3. Does having step-by-step solutions make it easier for you to understand?
4. How useful do you find online platforms for exam preparation?
5. Do you want to track your own learning progress?

Findings:

- ❖ Most students fear mathematics due to lack of conceptual clarity.
- ❖ Visual explanations significantly improve understanding.
- ❖ Teachers prefer structured content and progress tracking.
- ❖ Step-by-step solutions reduce confusion and errors.

Impact on SRS:

These findings shaped core functional requirements such as video tutorials, guided solutions, visualization modules, and performance dashboards.

4.1.2 Workshops**Purpose:**

To collaboratively identify functional priorities, learning flow, and interaction design among stakeholders.

Method:

- ❖ Conducted in a university classroom environment.
- ❖ **Participants:** 4 students, 2 teachers, and 1 faculty advisor.
- ❖ Brainstorming and feature prioritization techniques were used.

Outputs:

- ❖ Defined learning flow: Topic ➡ Video ➡ Visualization ➡ Practice ➡ Feedback
- ❖ Confirmed the need for difficulty levels (Easy, Medium, Hard)
- ❖ Introduced a new requirement: **Timer-based practice**

4.1.3 Focus Group Discussion

Purpose:

To explore collective expectations and compare student and teacher perspectives on learning effectiveness.

Setup:

- ❖ 8 participants (5 students and 3 teachers).
- ❖ One-hour moderated discussion.

Key Topics:

- ❖ Dashboard usability.
- ❖ Visualization effectiveness.
- ❖ Language preference.

Findings:

- ❖ Students prefer Bangla explanations for complex topics.
- ❖ Teachers want downloadable progress reports.
- ❖ Both groups value instant feedback.

Impact on SRS:

Resulted in bilingual UI support and report-generation features.

4.1.4 Observation

Purpose:

To understand real learning behavior and problem-solving patterns.

Process:

- ❖ Observed students solving mathematical problems manually.
- ❖ Monitored time spent per question and common error patterns.

Insights:

Students frequently skip intermediate steps, leading to mistakes.
Visualization improves formula retention.

Result:

Inspired enforced step-by-step solution flow and hint-based guidance in Calcure.

4.1.5 Questionnaires

Purpose:

To collect large-scale quantitative validation of requirements.

Respondents:

20 students and teachers

Sample Questions:**Questions for Students**

1. Which math topics do you find most difficult? (For example: Algebra, Geometry, Trigonometry)
2. When viewing a solution, do you prefer only the final answer, or a step-by-step explanation with animations?
3. What type of visualization or animation helps you understand math better?
4. On which device or platform would you like to use the software? (Desktop, Mobile, or Web)
5. What kind of easy-to-use and interactive features do you like most?
6. How important is time-saving or automatic solution generation for you?
7. Which features would encourage you to share the software with your friends or classmates?

Questions for Teachers

1. While teaching math in class, which topics feel the most difficult or time-consuming?
2. What type of animation or visualization do you think is necessary to help students understand better?
3. Which type of solution process is most effective for students? (Step-by-step, Interactive, or Quiz-based)
4. Should the software be used only by students, or also as a teaching aid in the classroom?
5. Which features would make teaching math easier for you?
6. Do you need any system to track student performance or results?
7. Do you want students to solve problems on their own, or should they be able to view guided solutions?

Findings:

- ❖ 92% prefer video-based explanations.
- ❖ 88% want instant feedback.
- ❖ 76% value progress tracking.

Impact:

Confirmed visualization, feedback mechanisms, and dashboards as core system features.

4.1.6 User Interface Analysis**Purpose:**

To evaluate usability and accessibility expectations.

Process:

- ❖ Low-fidelity paper prototypes.
- ❖ Click-flow observation and feedback.

Findings:

- ❖ Preference for simple layouts with large fonts.
- ❖ Request for dark mode for night study.
- ❖ Need for quick access to "Continue Learning".

Impact:

Shaped non-functional usability and accessibility requirements.

4.2 Sample of Requirement Collection

This section demonstrates how raw stakeholder inputs were transformed into formal system requirements.

STU-01: Students want visual explanations: The system shall provide interactive graphs and diagrams.

STU-02: Students want step-by-step help: The system shall display guided solutions progressively.

TCH-03: Teachers want tracking: The system shall generate performance reports.

ADM-04: System Admin wants control: The system shall support role-based access control.

PERF-05: Fast response: System pages shall load within two seconds under normal operating conditions.

All requirements are traceable to their respective elicitation sources.

4.3 Requirement Validation

Requirement validation ensures that the documented requirements for the **Calcure Online Mathematical Learning System** are correct, complete, consistent, and feasible. Although requirements may appear clear during documentation, ambiguities or conflicts may surface during later stages. Therefore, comprehensive validation is essential.

The validation process confirms that Calcure supports effective learning through guided solutions, visual explanations, video tutorials, and accurate progress tracking in a reliable and user-friendly manner.

4.3.1 Review the Requirements

A structured peer-review process was applied involving developers, academic supervisors, teachers, and domain experts. Review activities focused on:

- ❖ Functional and non-functional requirements.
- ❖ User workflows (Student, Teacher, Moderator, Admin).
- ❖ Learning effectiveness, usability, and accessibility.
- ❖ Content accuracy and role-based access control.

This process helped identify ambiguities, inconsistencies, and unrealistic expectations at an early stage, reducing the risk of major changes during development.

4.3.2 Test the Requirements

Requirement-based test cases were derived from documented requirements to validate system behavior. These tests ensured that:

- ❖ Each requirement is measurable and verifiable.
- ❖ System behavior is clearly defined for all user roles.
- ❖ Learning features function as intended.

Testing requirements prior to full implementation ensures development aligns with defined expectations.

4.3.3 Simulate the Requirements

Simulation-based validation was conducted using low-fidelity and interactive prototypes to visualize system behavior.

Simulation Activities Included:

- Clickable UI mockups for dashboards and learning modules
- Step-by-step problem-solving demonstrations
- Visualization and video navigation flows
- Progress tracking and feedback scenarios

Simulation enabled early detection of usability issues and learning-flow gaps, improving overall system quality before development.

5. Use Case Diagram

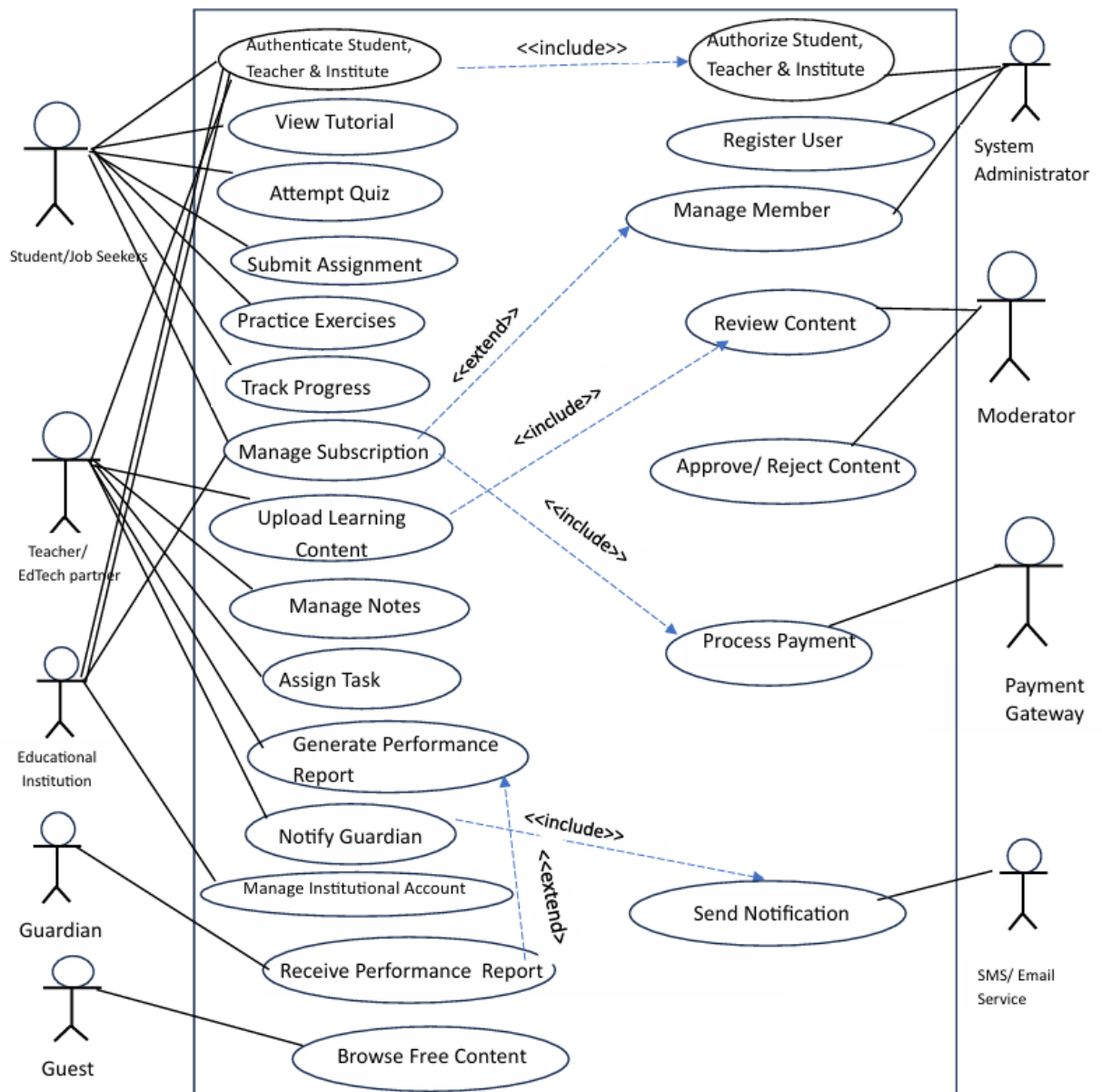


Figure 1: Use Case Diagram For **Calcure**

6. Use case Description

6.1 UC -01 : Authentication

Use Case No	UC -01
Use Case Name	Authentication
Goal	To verify the identity of a registered user and allow secure access to authorized features of the Calcure application.
Primary Actors	Student, Teacher, Educational Institution, Job candidate
Secondary Actors	Authentication System, Email Service/ SMS Gateway
Precondition	1. The user must be registered in the Calcure system. 2. The user account must be active and not suspended. 3. The system must be online.
Success End Condition	The user is successfully authenticated and granted access to authorized features.
Failed End Condition	The user fails to authenticate due to invalid credentials, inactive account, or system error.
Trigger	The user selects the " Login " option and submits login credentials.
Main Success Flow	1. The user opens the Calcure application. 2. The user selects the Login option. 3. The system displays the login form. 4. The user enters registered email/phone number and password. 5. The user submits the login request. 6. The system validates the credentials. 7. The system authenticates the user. 8. The system establishes a secure session. 9. The system redirects the user to the authorized dashboard.
Alternative Flows	A1: Invalid Credentials 6a. The system detects incorrect email/phone or password. 6b. The system displays an error message. 6c. The user retries authentication. A2: Unverified Account 6a. The system detects that the account is not verified. 6b. The system prompts the user to verify via email/SMS. 6c. The system resends the verification link/code if requested. A3: Account Suspended or Inactive 6a. The system detects that the account is suspended or inactive. 6b. The system displays an access denied message. 6c. The use case ends. A4: Forgot Password 4a. The user selects Forgot Password. 4b. The system initiates password recovery process.
Quality Attributes	1. Security The system shall encrypt all authentication data using HTTPS with TLS .

	2. Reliability The authentication mechanism shall correctly validate user credentials on maximum 5 login attempt. 3. Performance The system shall complete the login process and display the authentication result within 2 seconds under normal operating conditions. 4. Usability The login interface shall provide clear, user-friendly error messages for invalid credentials. 5. Availability The authentication service shall be operational 24/7 with minimal downtime
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6.2 UC -02 : Authorization

Use Case No	UC -02
Use Case Name	Authorization
Goal	To ensure that an authenticated user is granted access only to the system features and resources permitted by their assigned role (e.g., Student, Teacher, Guardian, Admin, Moderator).
Primary Actors	Student / Job Seeker, Teacher, Guardian, Admin, Moderator, Educational Institution, Project Owner
Secondary Actors	System Admin, Notification Service (optional, for security alerts)
Precondition	1. The user is already authenticated (valid login credentials verified). 2. The user has an assigned role in the Calcure System. 3. The Calcure System is online and operational.
Success End Condition	The system successfully verifies the user's role and permissions and he user is granted access to the authorized system functions.
Failed End Condition	The system fails to verify the user's permissions or access to the requested resource is denied.
Trigger	A logged-in user attempts to access a protected feature or resource (e.g., managing content, approving material, viewing reports).
Main Success Flow	1. The user requests access to a specific system feature. 2. The Calcure System retrieves the user's role and permission set. 3. The system checks the requested action against the authorized permissions. 4. The system confirms that the user has sufficient privileges. 5. The system grants access to the requested feature. 6. The user continues normal system interaction.
Alternative Flows	A1: Session Expired 1a. The system detects that the user session has expired. 1b. The system redirects the user to the login page.

	1c. The use case ends. A2: Unauthorized Access 2a. The system finds that the user does not have sufficient permissions. 2b. The system denies access and displays an error message. 2c. The use case ends. A3: Role Not Found 2a. The system fails to retrieve the user role information. 2b. The system denies access and logs the error. 2c. The use case ends.
Quality Attributes	Security: Prevents unauthorized access to sensitive system functions. Reliability: Ensures consistent permission checks across all modules. Performance: Authorization checks are performed with 5 sec. delay. Scalability: Supports at least 5 user roles and permission levels. Usability: Clear feedback is provided through SMS when access is denied

6.3 UC -03 : View Tutorial

Use Case No	UC -03
Use Case Name	View Tutorial
Goal	Allows a registered Student or Job Seeker to watch available video tutorials in the system learning for understanding mathematical topics.
Primary Actors	Student, Job Seekers
Secondary Actors	Admin (Authentication management system and subscription management system)
Precondition	- The user must be registered and logged in to the system. - The tutorial content must be approved by the moderator. - The user must have an active subscription for premium tutorials.
Success End Condition	The tutorial video is successfully played, and the user's learning progress is updated
Failed End Condition	The tutorial cannot be viewed due to authentication failure, missing subscription, or unavailable content.
Trigger	The user selects the " View Tutorial " option from the application after logging in.
Main Success Flow	- The user logs into the System. - The system authenticates the user successfully. - The user selects a subject (e.g., Algebra, Calculus). - The system displays a list of available tutorials for the selected subject. - The user selects a tutorial. - The system checks the user's subscription status. - The system grants access to the tutorial. - The tutorial video starts playing. - The system records the viewing activity for progress tracking

Alternative Flows	A1: User Not Logged In 1a. The user tries to view a tutorial without logging in. 1b. The system redirects the user to the login/registration page. 1c. The use case ends. A2: No Active Subscription 6a. The system detects that the user does not have an active subscription. 6b. The system prompts the user to subscribe. 6c. If the user subscribes, the main success flow resumes at step 7. 6d. If the user declines, the use case ends. A3: Tutorial Not Available 4a. No tutorial is available for the selected topic. 4b. The system shows a "No content available" message. 4c. The use case ends.
Quality Attributes	1. Usability – Tutorials should be easy to find and play with minimal user effort. 2. Performance – Video tutorials should start within 1 second loading time. 3. Security – Only authorized and subscribed users can access premium tutorials. 4. Availability – Tutorials should be accessible at all times except during maintenance. 5. Scalability – The system should support 10,000 users viewing tutorials simultaneously.

6.4 UC -04 : Attempt Quiz

Use Case No	UC -04
Use Case Name	Attempt Quiz
Goal	Allow a Student or Job Seeker to attempt quizzes related to a specific topic in order to evaluate their understanding and track learning progress
Primary Actors	Students , Job Seekers
Secondary Actors	Authentication System, Quiz Evaluation System
Precondition	1. The user must be registered and logged in. 2. The quiz must be created by a teacher and approved by a moderator. 3. The user must have an active subscription for premium quizzes .
Success End Condition	The quiz is successfully submitted, evaluated, and the score is stored in the system.
Failed End Condition	The quiz attempt fails due to authentication issues, subscription problems, or system errors.
Trigger	The user selects the " Attempt Quiz " option from the quiz section of the application.
Main Success Flow	1. The user logs into the Calcure application. 2. The system authenticates the user successfully.

	3. The user selects a subject and topic. 4. The system displays the available quizzes for the selected topic. 5. The user selects a quiz to attempt. 6. The system verifies the user's subscription status. 7. The system displays the quiz instructions and time limit. 8. The user starts the quiz. 9. The user answers all quiz questions. 10. The user submits the quiz. 11. The system evaluates the submitted answers. 12. The system displays the quiz score and correct answers. 13. The system stores the result for progress tracking.
Alternative Flows	A1: User Not Logged In 1a. The user tries to attempt a quiz without logging in. 1b. The system redirects the user to the login page. 1c. The use case ends. A2: No Active Subscription 6a. The system detects no active subscription. 6b. The system prompts the user to subscribe. 6c. If the user subscribes, the main flow resumes at step 7. 6d. If the user declines, the use case ends. A3: Quiz Time Expires 9a. The quiz time limit expires before submission. 9b. The system automatically submits the quiz. 9c. The system continues from step 11. A4: Quiz Not Available 4a. No quiz is available for the selected topic. 4b. The system displays an appropriate message. 4c. The use case ends.
Quality Attributes	1. Usability – Quiz interface should be clear and easy to navigate. 2. Performance – Quiz submission and evaluation should be processed within 24 hours. 3. Reliability – Answers should not be lost due to network or system failure. 4. Security – Only authorized users can attempt quizzes. 5. Accuracy – Quiz evaluation must be correct and consistent.

6.5 UC -05 : Submit Assignment

Use Case No	UC -05
Use Case Name	Submit Assignment
Goal	Allow a Student or Job Seeker to submit completed assignments assigned by teachers so that their performance can be evaluated.
Primary Actors	Students , Job Seekers
Secondary Actors	Authentication System, Assignment Management System
Precondition	1. The user must be registered and logged in .

	- The assignment must be created by a teacher and approved by a moderator (if required). - The user must have an active subscription (if the assignment is premium). - The assignment submission deadline has not passed.
Success End Condition	The assignment is successfully submitted, stored, and marked as "Submitted" in the system.
Failed End Condition	The assignment submission fails due to authentication failure, deadline expiration, invalid file format, or system error.
Trigger	The user selects the "Submit Assignment" option from the assignment section of the application.
Main Success Flow	- The user logs into the System - The system authenticates the user successfully. - The user navigates to the assignment section. - The system displays the list of active assignments. - The user selects an assignment to submit. - The system displays assignment details and submission guidelines. - The user uploads the completed assignment file or enters answers. - The user confirms the submission. - The system validates the submission format and deadline. - The system stores the assignment securely. - The system notifies the teacher about the submission. - The system updates the user's assignment status.
Alternative Flows	A1: User Not Logged In - The user attempts to submit an assignment without logging in. - The system redirects the user to the login page. - The use case ends. A2: Submission Deadline Passed - The system detects that the deadline has passed. - The system rejects the submission and displays a message. - The use case ends. A3: Invalid File Format - The system detects an unsupported file format. - The system prompts the user to upload a valid file. - The flow returns to step 7. A4: System Error During Upload - A system or network error occurs during upload. - The system displays an error message. - The user may retry submission.
Quality Attributes	Usability – Assignment submission should be simple and user-friendly. Reliability – Submitted assignments must not be lost or corrupted. Security – Only authorized users can submit assignments. Performance – Assignment upload should complete within 30 seconds .

	5. Integrity – Submitted content must remain unchanged after submission.
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6.6 UC -06 : Practice Exercise

Use Case No	UC -06
Use Case Name	Practice Exercise
Goal	Allows Student or Job Seeker to practice topic-wise mathematical exercises in order to improve skills and reinforce learning through immediate feedback.
Primary Actors	Students, Job Seekers.
Secondary Actors	Authentication System, Exercise Evaluation System.
Precondition	1. The user must be registered and logged in. 2. The practice exercises must be created by a teacher and approved by a moderator. 3. The user must have an active subscription for premium exercises (if applicable).
Success End Condition	The practice exercise is completed successfully, feedback is shown, and the user's progress is updated.
Failed End Condition	The practice exercise cannot be completed due to authentication failure, subscription issues, or system errors.
Trigger	The user selects the " Practice Exercise " option from the practice section of the application.
Main Success Flow	1. The user logs into the Calcure application. 2. The system authenticates the user successfully. 3. The user selects a subject and topic. 4. The system displays available practice exercises. 5. The user selects an exercise set. 6. The system verifies the user's subscription status. 7. The system displays the exercise questions. 8. The user solves the exercises one by one. 9. The user submits answers for evaluation. 10. The system evaluates the answers instantly. 11. The system displays correct answers and explanations. 12. The system records the practice activity for progress tracking.
Alternative Flows	A1: User Not Logged In 1a. The user attempts to access practice exercises without logging in. 1b. The system redirects the user to the login page. 1c. The use case ends. A2: No Active Subscription 6a. The system detects that the user has no active subscription. 6b. The system prompts the user to subscribe. 6c. If the user subscribes, the main flow resumes at step 7.

	6d. If the user declines, the use case ends. A3: Exercise Not Available 4a. No practice exercise is available for the selected topic. 4b. The system displays an appropriate message. 4c. The use case ends.
Quality Attributes	1. Usability – Practice interface should be simple and interactive(GUI). 2. Performance – Answers should be evaluated within 1 second. 3. Reliability – Practice progress should not be lost due to failures. 4. Security – Only authorized users can access practice exercises. 5. Scalability – The system should support at least 10,000 users practicing simultaneously.

6.7 UC -07 : Track Progress

Use Case No	UC -07
Use Case Name	Track Progress
Goal	Allows a Student or Job Seeker to view and monitor their learning progress, performance statistics, and activity history within the System.
Primary Actors	Student , Job Seekers
Secondary Actors	Authentication System , Progress Tracking System
Precondition	1. The user must be registered and logged in . 2. The user must have completed at least one activity (tutorial view, quiz, assignment, or practice exercise).
Success End Condition	The user successfully views an up-to-date summary of their learning progress.
Failed End Condition	Progress tracking fails due to authentication issues, missing data, or system errors.
Trigger	The user selects the " Track Progress " option from the dashboard or profile section.
Main Success Flow	1. The user logs into the System. 2. The system authenticates the user successfully. 3. The user navigates to the Track Progress section. 4. The system retrieves the user's learning data. 5. The system displays progress details, including: 6. Tutorials viewed 7. Quiz scores 8. Assignment submission status 9. Practice exercise performance 10. The system presents visual summaries (charts/percentages). 11. The user reviews progress and performance insights.

Alternative Flows	A1: User Not Logged In 1a. The user attempts to access progress tracking without logging in. 1b. The system redirects the user to the login page. 1c. The use case ends. A2: No Learning Activity Found 4a. The system finds no learning activity for the user. 4b. The system displays a message such as "No progress data available yet." 4c. The use case ends. A3: System Data Retrieval Error 4a. The system fails to retrieve progress data. 4b. The system displays an error message. 4c. The user may retry later.
Quality Attributes	1. Usability – Progress information should be easy to understand and visually clear. 2. Accuracy – Displayed progress data must be correct and up to date. 3. Performance – Progress data should load within 5 second response time. 4. Reliability – Progress records must be stored in database and retrieved within milisecond. 5. Security – Only the respective user can view their progress data.

6.8 UC -08 : Manage Subscription

Use Case No	UC -08
Use Case Name	Manage Subscription
Goal	To allow Students, Job Seekers, and Educational Institutions to subscribe, renew, upgrade, downgrade, or cancel their subscriptions in order to access Calcure's premium services .
Primary Actors	Student, Job Candidate , Educational Institution
Secondary Actors	Payment Gateway , Subscription Management System
Precondition	1. The actor must be registered and logged in. 2. Subscription plans (individual and institutional) must be available. 3. Institutional accounts must be approved by the admin (for institutions).
Success End Condition	The subscription (individual or institutional) is successfully updated and activated in the system.
Failed End Condition	Subscription management fails due to payment issues, approval rejection, or system errors.
Trigger	The actor selects the " Manage Subscription " option from the dashboard or account settings.

Main Flow	Success 1. The actor logs into the Calcure application. 2. The system authenticates the actor successfully. 3. The actor navigates to the Manage Subscription section. 4. The system displays: ❖ Current subscription status ❖ Available subscription plans (individual or institutional) 5. The actor selects a subscription action: ❖ Subscribe ❖ Renew ❖ Upgrade ❖ Downgrade ❖ Cancel 6. If payment is required, the system redirects to the payment gateway. 7. The actor completes the payment process. 8. The payment gateway confirms successful payment. 9. The system updates the subscription status. 10. The system activates or updates access privileges: • Individual access for students/job seekers • Multi-user institutional access for educational institutions 11. The system displays a confirmation message.
Alternative Flows	A1: Payment Failure 8a. The payment gateway reports payment failure. 8b. The system displays an error message. 8c. The actor may retry payment or choose another method. A2: Institutional Subscription Approval Required 5a. An educational institution selects an institutional subscription. 5b. The system sends the request for admin approval. 5c. The subscription remains pending until approval. A3: Subscription Cancellation 5a. The actor selects cancel subscription. 5b. The system confirms cancellation. 5c. The subscription remains active until the billing period ends.
Quality Attributes	1. Usability – Subscription management should be simple for both individuals and institutions. 2. Security – Payment and institutional data must be handled in encrypted way. 3. Reliability – Subscription updates must be consistently stored in database. 4. Scalability – The system should support institutional subscriptions with 1000 member access.

	5. Availability – Subscription services should be accessible at 24/7.
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6.9 UC -09 : Upload Content

Use Case No	UC -09
Use Case Name	Upload Content
Goal	To allow a Teacher to upload educational content (video tutorials, step-by-step solutions, notes, quizzes, and assignments) to the Calcure platform for student learning.
Primary Actors	Teacher
Secondary Actors	Content Management System, Moderator
Precondition	1. The teacher must be registered and logged in. 2. The teacher's account must be verified and approved by the admin. 3. The content must comply with Calcure's content guidelines.
Success End Condition	The content is successfully uploaded and marked as Pending Approval for moderator review.
Failed End Condition	The content upload fails due to validation errors, system failures, or non-compliance with guidelines.
Trigger	The teacher selects the " Upload Content " option from the teacher dashboard.
Main Success Flow	1. The teacher logs into the Calcure application. 2. The system authenticates the teacher successfully. 3. The teacher navigates to the Upload Content section. 4. The system displays content upload options: ❖ Video tutorial ❖ Step-by-step solution ❖ Notes ❖ Quiz ❖ Assignment 5. The teacher selects the content type. 6. The teacher enters content details (title, subject, topic, description). 7. The teacher uploads the content file or inputs content data. 8. The teacher submits the content for review. 9. The system stores the content with "Pending Approval" status. 10. The system notifies the moderator for review.
Alternative Flows	A1: Invalid Content Format 7a. The uploaded file format is not supported. 7b. The system displays an error message. 7c. The teacher uploads a valid format and resumes from step 7. A2: Incomplete Content Information 6a. Required content details are missing.

	6b. The system prompts the teacher to complete all mandatory fields. 6c. The flow resumes from step 6. A3: Upload Failure 7a. A network or system error occurs during upload. 7b. The system displays an error message. 7c. The teacher may retry uploading.
Quality Attributes	1. Usability – Content upload interface should be simple and intuitive. 2. Security – Only authorized teachers can upload content. 3. Reliability – Uploaded content must not be lost or corrupted. 4. Performance – Content upload should complete within acceptable time. 5. Maintainability – Uploaded content should be easy to update or remove.

6.10 UC -10 : Manage Notes

Use Case No	UC -10
Use Case Name	Manage Notes
Goal	To allow a Teacher to create, update, and delete lecture notes related to specific mathematical topics so that students can access structured learning materials.
Primary Actors	Teacher
Secondary Actors	Content Management System, Moderator
Precondition	1. The teacher must be registered and logged in . 2. The teacher's account must be approved by the admin . 3. The notes must follow Calcure's content guidelines.
Success End Condition	The notes are successfully created, updated, or deleted and sent for moderator approval.
Failed End Condition	Notes management fails due to validation errors, authorization issues, or system failures.
Trigger	The teacher selects the " Manage Notes " option from the teacher dashboard.
Main Success Flow	1. The teacher logs into the Calcure application. 2. The system authenticates the teacher successfully. 3. The teacher navigates to the Manage Notes section. 4. The system displays existing notes (if any). 5. The teacher selects an action: ❖ Create new notes ❖ Edit existing notes ❖ Delete notes

	6. The teacher enters or modifies note content (text, images, formulas). 7. The teacher saves the notes. 8. The system stores the notes with "Pending Approval" status. 9. The system notifies the moderator for review.
Alternative Flows	A1: Invalid or Incomplete Note Content 6a. Required note information is missing. 6b. The system displays a validation error message. 6c. The teacher completes the required fields and resumes from step 6. A2: Delete Notes 5a. The teacher selects delete note option. 5b. The system asks for confirmation. 5c. The system deletes the note and updates the list. A3: System Error While Saving 7a. A system error occurs during note saving. 7b. The system displays an error message. 7c. The teacher may retry later.
Quality Attributes	1. Usability – Notes should be in PDF format for easy to create and edit. 2. Security – Only authorized teachers can manage notes. 3. Reliability – Notes data must not be lost or corrupted. 4. Maintainability – Notes should be easy to update or remove. 5. Consistency – Notes should maintain consistent formatting and structure.

6.11 UC -11 : Assign Task

Use Case No	UC -11
Use Case Name	Assign Task
Goal	To allow a Teacher to assign tasks (assignments or practice sets) to students so that they can practice, apply concepts, and be evaluated.
Primary Actors	Teacher
Secondary Actors	Notification System
Precondition	1. The teacher must be registered and logged in . 2. The teacher account must be approved by the admin . 3. The related content (topic, notes, tutorials) must already exist.
Success End Condition	The task is successfully assigned and visible to the selected students.
Failed End Condition	Task assignment fails due to validation errors, authorization issues, or system failures.

Trigger	The teacher selects the “ Assign Task ” option from the teacher dashboard.
Main Success Flow	1. The teacher logs into the Calcure application. 2. The system authenticates the teacher successfully. 3. The teacher navigates to the Assign Task section. 4. The system displays task creation options (assignment / practice set). 5. The teacher selects the task type. 6. The teacher enters task details: ❖ Title ❖ Subject and topic ❖ Instructions ❖ Due date ❖ Marks / difficulty level 7. The teacher selects target students or groups. 8. The teacher submits the task. 9. The system stores the task. 10. The system notifies the assigned students. 11. The task becomes visible in students’ task list.
Alternative Flows	A1: Missing Task Information 6a. Required task details are missing. 6b. The system prompts the teacher to complete all mandatory fields. 6c. The flow resumes from step 6. A2: Invalid Due Date 6a. The due date is earlier than the current date. 6b. The system displays an error message. 6c. The teacher corrects the date and continues. A3: System Error During Assignment 9a. A system error occurs while saving the task. 9b. The system displays an error message. 9c. The teacher may retry later.
Quality Attributes	1. Usability – Task assignment should be simple and quick. 2. Reliability – Assigned tasks must not be duplicated. 3. Security – Only authorized teachers can assign tasks. 4. Performance – Tasks should be assigned and notified through SMS. 5. Scalability – The system should support assigning tasks to student groups consist of at least 50 students

6.12 UC -12 : Notify Guardian

Use Case No	UC -12
Use Case Name	Notify Guardian

Goal	To notify a Guardian about a student's academic performance, progress, or important updates through SMS or email .
Primary Actors	Teacher, Admin
Secondary Actors	Guardian, SMS Gateway, Email Service
Precondition	1. The student must be registered in the system. 2. Guardian contact information (phone number or email) must be available. 3. Performance data or notification content must exist.
Success End Condition	The guardian is successfully notified, and the notification is logged in the system.
Failed End Condition	Notification fails due to missing contact details, delivery issues, or system errors.
Trigger	The teacher or admin selects the " Notify Guardian " option after generating a performance report or important update.
Main Success Flow	1. The teacher/admin logs into the Calcure application. 2. The system authenticates the actor successfully. 3. The actor selects a student or group of students. 4. The system retrieves the guardian contact details. 5. The actor selects the notification type (performance report / alert). 6. The actor chooses the delivery method (SMS or email). 7. The system formats the notification message. 8. The system sends the notification via the selected channel. 9. The system records the notification status. 10. The guardian receives the notification.
Alternative Flows	A1: Guardian Contact Not Available 4a. The system finds no guardian contact details. 4b. The system displays an error message. 4c. The use case ends. A2: Notification Delivery Failure 8a. SMS/email delivery fails. 8b. The system logs the failure. 8c. The actor may retry or choose another delivery method. A3: Actor Cancels Notification 6a. The actor cancels the notification before sending. 6b. The system aborts the operation.
Quality Attributes	1. Reliability – Notifications must be delivered to exact guardian. 2. Timeliness – Notifications should be sent within 30 min of report generation. 3. Security – Guardian information only accessible by the guardian and authenticated Teacher . 4. Availability – Notification services should be accessible 24/7.

6.13 UC -13 : Generate Performance Report

Use Case No	UC -13
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Use Case Name	Generate Performance Report
Goal	To generate a comprehensive performance report for a student based on quizzes, assignments, practice exercises, and tutorial activities, so that academic progress can be evaluated and shared.
Primary Actors	Teacher, Admin
Secondary Actors	Performance Analysis System, Report Management System
Precondition	1. The student must be registered in the system. 2. The student must have completed at least one learning activity (quiz, assignment, practice exercise, or tutorial). 3. The teacher/admin must be authorized to generate reports.
Success End Condition	A complete and accurate performance report is successfully generated and stored in the system.
Failed End Condition	The report cannot be generated due to missing data, authorization issues, or system errors.
Trigger	The teacher or admin selects the "Generate Performance Report" option from the reporting section.
Main Success Flow	1. The teacher/admin logs into the Calcure application. 2. The system authenticates the actor successfully. 3. The actor selects a student or group of students. 4. The system retrieves the student's learning data. 5. The system analyzes performance data, including: ❖ Quiz scores ❖ Assignment results ❖ Practice exercise outcomes ❖ Tutorial activity 6. The system generates a structured performance report. 7. The system displays the report to the actor. 8. The actor may choose to download or share the report. 9. The system stores the report for future reference.
Alternative Flows	A1: No Performance Data Available 4a. The system finds no learning data for the selected student. 4b. The system displays a message indicating no data is available. 4c. The use case ends. A2: Report Generation Failure 6a. The system encounters an error during report generation. 6b. The system displays an error message. 6c. The actor may retry later.
Quality Attributes	1. Accuracy – Reports must correctly reflect student performance. 2. Performance – Report generation should complete within 7 days. 3. Reliability – Reports must be generated on student's actual performance and stored. 4. Security – Only respective student ,guardian and teachers can view reports.

	5. Maintainability – Report formats should be in PDF .
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6.14 UC -14 : Receive Performance Report

Use Case No	UC -14
Use Case Name	Receive Performance Report
Goal	To allow a Student or Guardian to receive and view the student's academic performance report generated by the system.
Primary Actors	Student, Guardian
Secondary Actors	Notification System(SMS/Email)
Precondition	1. A performance report must be generated by the teacher or admin. 2. The student must be registered in the system. 3. Guardian contact details must be available (for guardian access).
Success End Condition	The performance report is successfully received and viewed by the student or guardian.
Failed End Condition	The report cannot be received due to delivery failure, missing data, or system errors.
Trigger	The system sends a performance report after it is generated by a teacher or admin, or the student selects " View Performance Report ".
Main Success Flow	1. The system generates the performance report. 2. The system sends a notification to the student or guardian via SMS or email. 3. The student/guardian receives the notification. 4. The student/guardian opens the report link or logs into the system. 5. The system displays the performance report. 6. The student/guardian reviews the report details.
Alternative Flows	A1: Guardian Not Registered 2a. Guardian is not registered in the system. 2b. The system sends the report only to the student. 2c. The use case ends. A2: Notification Delivery Failure 2a. SMS/email delivery fails. 2b. The system logs the failure. 2c. The user may access the report directly from the system. A3: User Not Logged In (Student View) 4a. Student attempts to view the report without logging in. 4b. The system redirects to the login page. 4c. The flow resumes after successful login.
Quality Attributes	1. Availability – Reports should be accessible 24/7. 2. Security – Only teacher, respective students and their guardian can view the report.

	3. Reliability – Reports must be delivered accurately. 4. Usability – Reports should be written in both Bangla and English language . 5. Confidentiality – Student performance data must remain private.
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6.15 UC -15 : Manage Institutional Account

Use Case No	UC -15
Use Case Name	Manage Institutional Account
Goal	To allow an Educational Institution to manage its institutional account on the Calcure platform, including user management, subscription status, and access privileges for students and teachers.
Primary Actors	Educational Institution (School / College / Training Center)
Secondary Actors	Admin, Account Manage System, Subscription Management System
Precondition	1. The educational institution must be registered and approved by the admin. 2. The institutional account must be active. 3. The authorized institutional representative must be logged in.
Success End Condition	The institutional account is successfully updated and reflects the latest changes.
Failed End Condition	Institutional account management fails due to authorization issues, subscription problems, or system errors.
Trigger	The educational institution selects the " Manage Institutional Account " option from its dashboard.
Main Success Flow	1. The institutional representative logs into the Calcure application. 2. The system authenticates the institutional account. 3. The representative navigates to the Manage Institutional Account section. 4. The system displays institutional account details, including: ❖ Subscription status ❖ Number of active users ❖ Account validity period 5. The representative selects an account management option: ❖ Add or remove students ❖ Add or remove teachers ❖ Update institutional profile information ❖ View subscription details 6. The representative submits the requested changes. 7. The system validates the changes. 8. The system updates the institutional account records. 9. The system confirms successful account management.

Alternative Flows	A1: Subscription Expired 4a. The system detects that the institutional subscription has expired. 4b. The system prompts the institution to renew the subscription. 4c. The use case ends until renewal is completed. A2: Unauthorized Access 2a. The user attempting access is not authorized. 2b. The system denies access and displays an error message. 2c. The use case ends. A3: Invalid User Limit 6a. The institution attempts to add users beyond the allowed limit. 6b. The system displays a warning message. 6c. The representative may upgrade the subscription or revise the request.
Quality Attributes	1. Security – Only authorized institutional representatives can manage the account. 2. Scalability – The system must support at least 50 institutional users. 3. Reliability – Institutional data must be stored in an encrypted way and updated after 30 days of duration. 4. Usability – Account management should be simple and well organized. 5. Auditability – All institutional account changes should be logged.

6.16 UC -16 : Browse Free Content

Use Case No	UC -16
Use Case Name	Browse Free Content
Goal	To allow a Guest (unregistered user) to explore free tutorials, notes, exercises, and other non-premium learning materials in the Calcure application without requiring registration or subscription.
Primary Actors	Guest
Secondary Actors	Content Management System
Precondition	1. The guest must have access to the Calcure application interface (mobile or web). 2. Free content must be available and published in the system.
Success End Condition	The guest successfully browses and views free content without registration.
Failed End Condition	The guest cannot view any free content because none is available
Trigger	A guest selects the " Browse Free Content " option from the Calcure homepage or navigation menu.
Main Success Flow	1. The guest launches the Calcure application or visits the Calcure homepage. 2. The guest selects "Browse Free Content."

	3. The system retrieves a list of free tutorials, notes, practice exercises, and videos. 4. The system displays the list of free content categorized by subject and topic. 5. The guest selects a content item (e.g., tutorial, notes). 6. The system checks that the selected content is free. 7. The system displays the selected content to the guest. 8. The guest views the free content or navigates to another free item.
Alternative Flows	A1: No Free Content Available 3a. The system finds no free content available. 3b. The system displays a message: "No free content available." 3c. The use case ends. A2: Guest Tries to Access Premium Content 6a. The guest selects a premium/paid item. 6b. The system shows a prompt to register or subscribe. 6c. The guest may either: ❖ Choose to register/subscribe (redirect to registration/payment), or ❖ Return to browsing free items.
Quality Attributes	1. Usability – Free content must be easy to browse and access with a clear GUI. 2. Performance – Content listing and loading should occur within 5 seconds response time. 3. Scalability – The system should support 50 number of guests browsing simultaneously. 4. Availability – Free content should be available at all times except during maintenance. 5. Security – Guests must not access premium features unless registered and subscribed.

6.17 UC -17 : Register User

Use Case No	UC -17
Use Case Name	Register User
Goal	To allow a new user to create an account in the Calcure application so that they can access learning features and services.
Primary Actors	Student, Job Candidate, Teacher, Educational Institution
Secondary Actors	Admin, Email Service/ SMS Gateway
Precondition	1. The user must not already have a registered account. 2. The Calcure system must be available.
Success End Condition	The user account is successfully created and ready for login (or pending admin approval).

Failed End Condition	Registration fails due to validation errors, duplicate accounts, or system failure.
Trigger	The user selects the " Register " option from the Calcure homepage or login screen.
Main Success Flow	7. The user opens the Calcure application. 8. The user selects the Register option. 9. The system displays the registration form. 10. The user selects the user type (Student / Job Seeker / Teacher / Institution). 11. The user enters required information: ❖ Name ❖ Email address ❖ Phone number ❖ Password 12. Educational details (if applicable) 13. The user submits the registration form. 14. The system validates the provided information. 15. The system creates a new user account. 16. The system sends a verification message via email or SMS. 17. The system displays a successful registration confirmation.
Alternative Flows	A1: Invalid or Missing Information 7a. The system detects invalid or incomplete data. 7b. The system displays validation errors. 7c. The user corrects the information and resubmits. A2: User Already Registered 7a. The system detects that the email or phone number already exists. 7b. The system displays a "User already registered" message. 7c. The use case ends. A3: Teacher / Institution Requires Admin Approval 8a. The system marks the account as Pending Approval. 8b. The admin is notified. 8c. The account becomes active after admin approval.
Quality Attributes	1. Usability – Registration should be completed using user's identity, Phone number ,Email and other official information . 2. Security – User credentials should be accessible by users only. 3. Reliability – User data must not be lost during registration. 4. Performance – Registration should complete within 5 min. 5. Scalability – The system should handle at least 500 number of registrations at a time.

6.18 UC -18 : Manage Member

Use Case No	UC -18
Use Case Name	Manage Member

Goal	To allow the Admin to add, update, activate, suspend, or remove members in the Calcure System.
Primary Actors	Admin
Secondary Actors	System Audit Service, Notification Service(SMS/ Email)
Precondition	1. Admin is authenticated and authorized. 2. Member records exist or are ready to be created.
Success End Condition	
Failed End Condition	
Trigger	Admin selects the Manage Member option from the admin dashboard.
Main Success Flow	1. Admin accesses the Manage Member feature. 2. The system displays the list of registered members. 3. Admin selects an action (add, edit, activate, suspend, or delete member). 4. Admin enters or modifies member information. 5. The system validates the provided data. 6. The system updates the member record. 7. The system confirms successful completion.
Alternative Flows	A1: Invalid Member Information 4a. The system detects invalid or incomplete member data. 4b. The system displays validation error messages. 4c. The Admin corrects the data and resumes the main flow. A2: Member Not Found 3a. The system cannot locate the selected member. 3b. The system displays a "Member Not Found" message. 3c. The use case ends. A3: System Failure 6a. A system error occurs while saving member information. 6b. The system aborts the operation and logs the error. 6c. The use case ends.
Quality Attributes	1. Security: Only Admin can manage members. 2. Reliability: Ensures consistent and accurate member data. 3. Usability: Simple and clear member management interface. 4. Performance: Response within 3 seconds for member operations.

6.19 UC -19 : Review Content

Use Case No	UC -19
Use Case Name	Review Content
Goal	To allow the Moderator to review submitted educational content and decide whether it meets the quality and policy standards of the Calcure System.
Primary Actors	Moderator

Secondary Actors	Teacher, EdTech Partner, System Audit Service, Notification Service (SMS/Email)
Precondition	1. Moderator is authenticated and authorized. 2. Content has been submitted by a Teacher or EdTech Partner.
Success End Condition	The content is successfully reviewed and marked as approved or rejected in the system.
Failed End Condition	The content review process fails and the content status remains unchanged.
Trigger	Moderator selects pending content from the review queue.
Main Success Flow	1. Moderator accesses the Review Content feature. 2. The system displays a list of pending content submissions. 3. Moderator selects a content item for review. 4. The system displays the content details. 5. Moderator evaluates the content quality and compliance. 6. Moderator submits the review decision. 7. The system updates the content status. 8. The system confirms the review completion.
Alternative Flows	A1: Content Does Not Meet Standards 5a. The moderator determines that the content violates quality or policy rules. 5b. The moderator marks the content for rejection. 5c. The flow proceeds to the approval/rejection decision. A2: Content Not Available 2a. The selected content is no longer available in the system. 2b. The system displays an error message. 2c. The use case ends. A3: System Error 6a. A system error occurs while saving the review decision. 6b. The system logs the error and displays a failure message. 6c. The use case ends.
Quality Attributes	1. Security: Only authorized moderators can review content. 2. Reliability: Ensures consistent review decisions. 3. Usability: Simple and intuitive content review interface. 4. Performance: within 30 min the content should be accessed for review.

6.20 UC -20 : Approve/ Reject Content

Use Case No	UC -20
Use Case Name	Approve/ Reject Content
Goal	To allow the Moderator to approve high-quality content or reject inappropriate or low-quality content before it is published in the Calcure System.
Primary Actors	Moderator

Secondary Actors	Teacher, EdTech Partner, Notification Service (SMS/ Email)
Precondition	3. Moderator is authenticated and authorized. 4. Content has already been reviewed and is pending approval
Success End Condition	The content is successfully approved or rejected and its status is updated in the system.
Failed End Condition	The approval or rejection process fails and the content status remains unchanged.
Trigger	Moderator selects a reviewed content item for approval or rejection.
Main Success Flow	1. Moderator accesses the Approve / Reject Content function. 2. The system displays reviewed content awaiting decision. 3. Moderator selects a content item. 4. Moderator chooses Approve or Reject. 5. Moderator optionally adds comments. 6. The system validates the action. 7. The system updates the content status. 8. The system confirms successful completion.
Alternative Flows	A1: Content Rejected 4a. The moderator selects the Reject option. 4b. The system records rejection comments and updates status. 4c. The system notifies the content provider. 4d. The use case ends. A2: Unauthorized Attempt 1a. The system detects that the user is not authorized. 1b. The system denies access and logs the attempt. 1c. The use case ends. A3: System Failure 7a. A system error occurs while updating content status. 7b. The system aborts the operation and logs the error. 7c. The use case ends.
Quality Attributes	7. Security: Only Moderators can approve or reject content. 8. Reliability: Ensures accurate content status updates. 9. Usability: Clear approval and rejection options. 10. Performance: Processing of approval decisions within 5 min of content review.

6.21 UC -21 : Process Payment

Use Case No	UC -21
Use Case Name	Process payment
Goal	To allow users or institutions to make payments for subscriptions, memberships, or services in the Calcure System.
Primary Actors	Student, Admin, Educational Institution
Secondary Actors	Payment Gateway, Notification Service
Precondition	1. User or institution is authenticated and authorized. 2. A valid payable service or subscription is selected.

Success End Condition	Payment is successfully processed and the transaction is recorded in the system.
Failed End Condition	Payment processing fails and no transaction is recorded.
Trigger	The user or institution selects the Pay Now option for a service or subscription.
Main Success Flow	1. The actor initiates the Process Payment function. 2. The system displays payable amount and payment options. 3. The actor selects a payment method. 4. The system redirects the request to the payment gateway. 5. The payment gateway processes the transaction. 6. The payment gateway returns a success response. 7. The system records the transaction details. 8. The system confirms successful payment to the actor.
Alternative Flows	A1: Payment Failed 5a. The payment gateway reports a failed transaction. 5b. The system displays a payment failure message. 5c. The use case ends. A2: Payment Cancelled by User 3a. The actor cancels the payment process. 3b. The system stops the transaction. 3c. The use case ends. A3: Payment Gateway Unavailable 4a. The payment gateway is unreachable. 4b. The system displays a service unavailable message. 4c. The use case ends. A4: Invalid Payment Details 3a. The actor enters invalid payment information. 3b. The system displays validation errors. 3c. The actor corrects the information and resumes the main flow.
Quality Attributes	1. Security: Ensures secure and encrypted payment processing. 2. Reliability: Prevents duplicate or inconsistent transactions. 3. Performance: Processes payments with 5 seconds delay. 4. Availability: Handles gateway downtime gracefully. 5. Usability: Provides clear feedback through SMS or Email during payment processing.

6.22 UC -22 : Send Notification

Use Case No	UC -22
Use Case Name	Send Notification
Goal	To notify users or institutions about important system events such as approvals, rejections, payments, membership updates, or system alerts.
Primary Actors	System
Secondary Actors	User(Student, Teacher, Admin, Moderate, Institution), Notification Service
Precondition	1. A valid system event has occurred. 2. Recipient user or institution exists in the system.

	3. Notification service is available.
Success End Condition	The notification is successfully delivered to the intended recipient.
Failed End Condition	The notification fails to be delivered and the event is logged.
Trigger	A system event requires notifying a user or institution.
Main Success Flow	1. A system event triggers the Send Notification use case. 2. The system identifies the notification recipients. 3. The system selects the notification type (email, SMS, push). 4. The system generates the notification content. 5. The system sends the notification via the notification service. 6. The notification service confirms successful delivery. 7. The system logs the notification status.
Alternative Flows	A1: Notification Service Unavailable 5a. The notification service is unreachable. 5b. The system records the delivery failure. 5c. The use case ends. A2: Invalid Recipient Information 2a. The system detects missing or invalid recipient details. 2b. The system aborts notification delivery. 2c. The use case ends. A3: Delivery Failure 6a. The notification service reports unsuccessful delivery. 6b. The system logs the failure for retry or review. 6c. The use case ends.
Quality Attributes	1. Reliability: Ensures consistent message delivery. 2. Availability: Supports at least 3 notification channels. 3. Performance: Sends notifications within 5 seconds delay. 4. Scalability: Handles high volumes of notifications. 5. Usability: Delivers clear and understandable messages written both in Bengali and English .

7. Activity Diagrams

7.1 Activity Diagram of Authentication

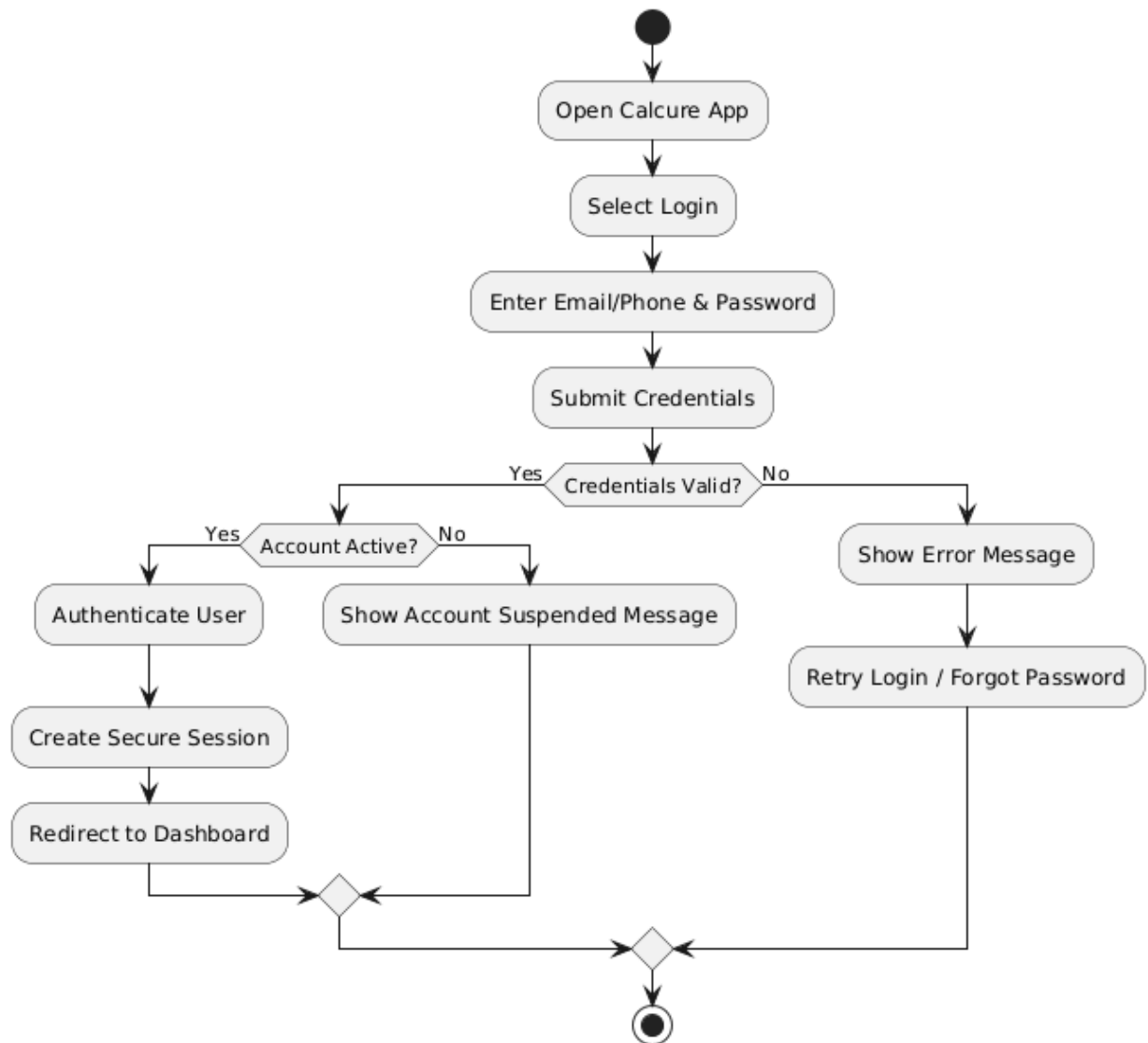


Figure 2: Activity diagram for Authentication

7.2 Activity Diagram of Authorization

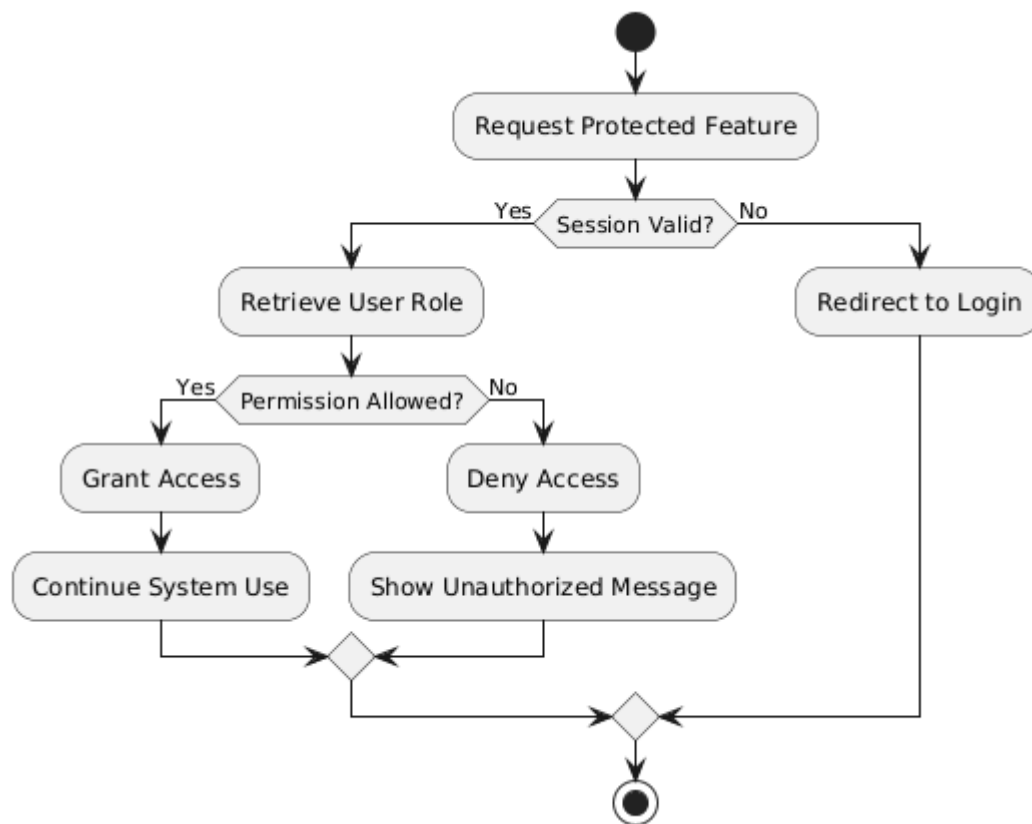


Figure 3: Activity diagram for Authorization

7.3 Activity Diagram of View Tutorial

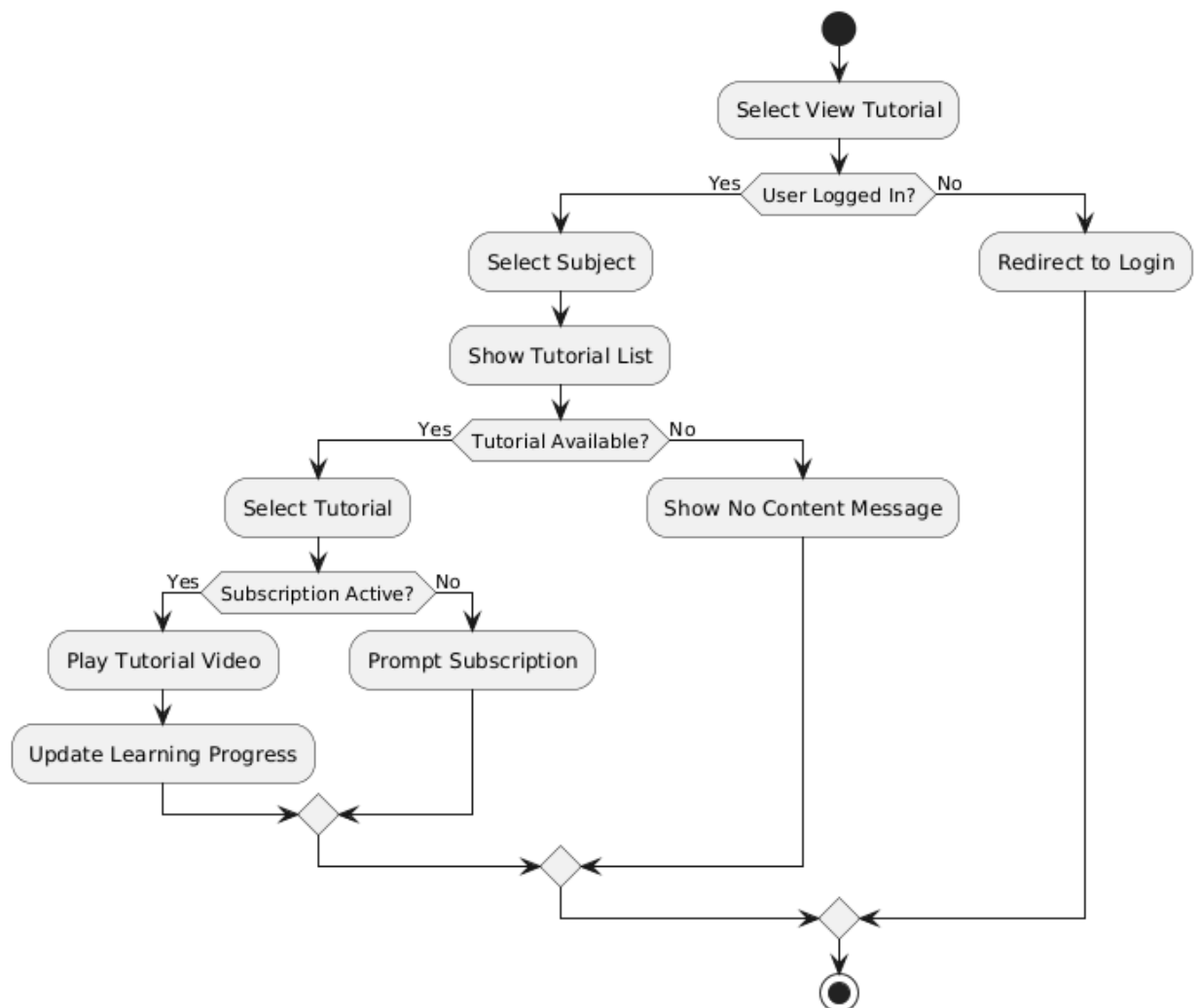


Figure 4: Activity diagram for View Tutorial

7.4 Activity Diagram of Attempt Quiz

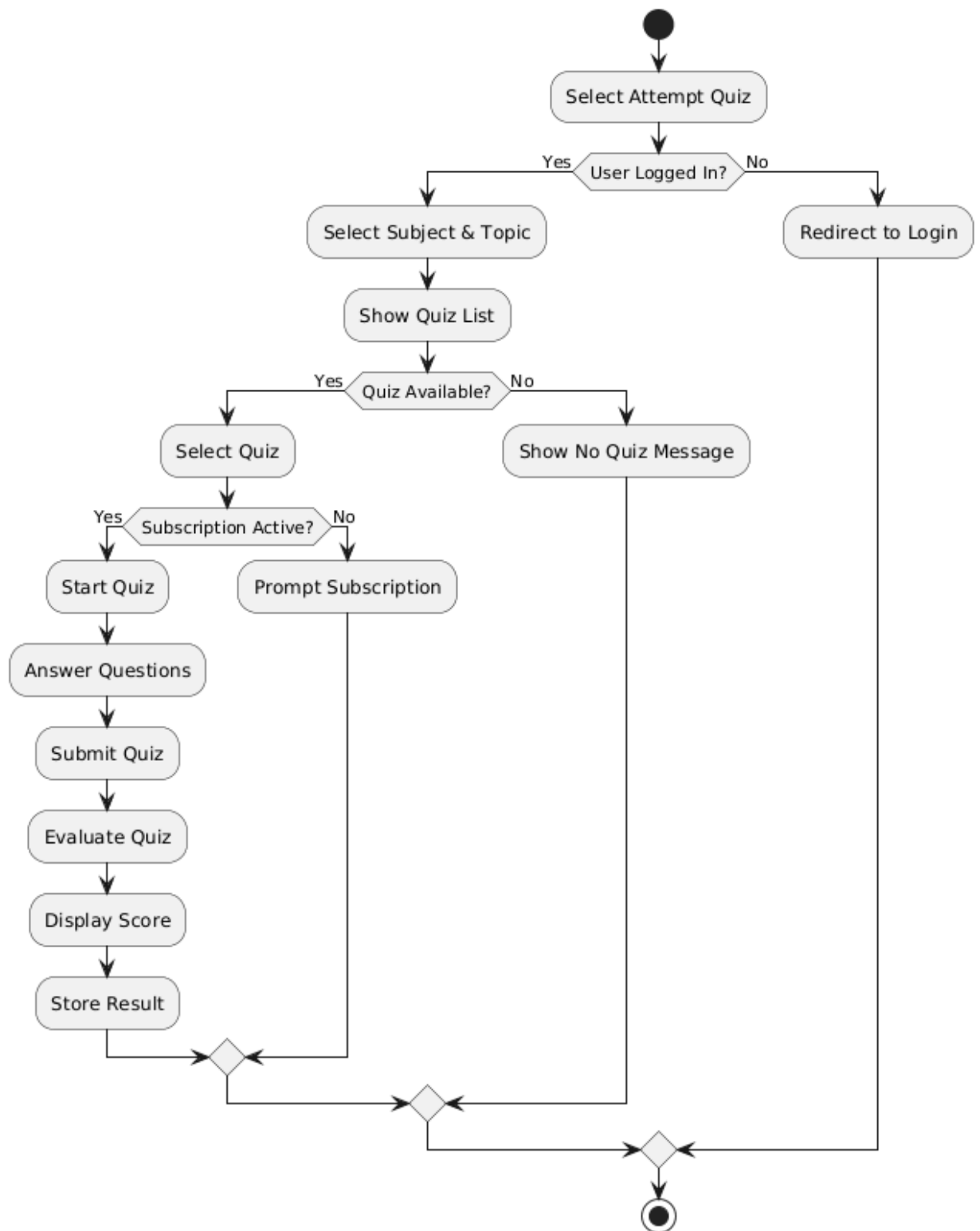


Figure 5: Activity Diagram for Attempt Quiz

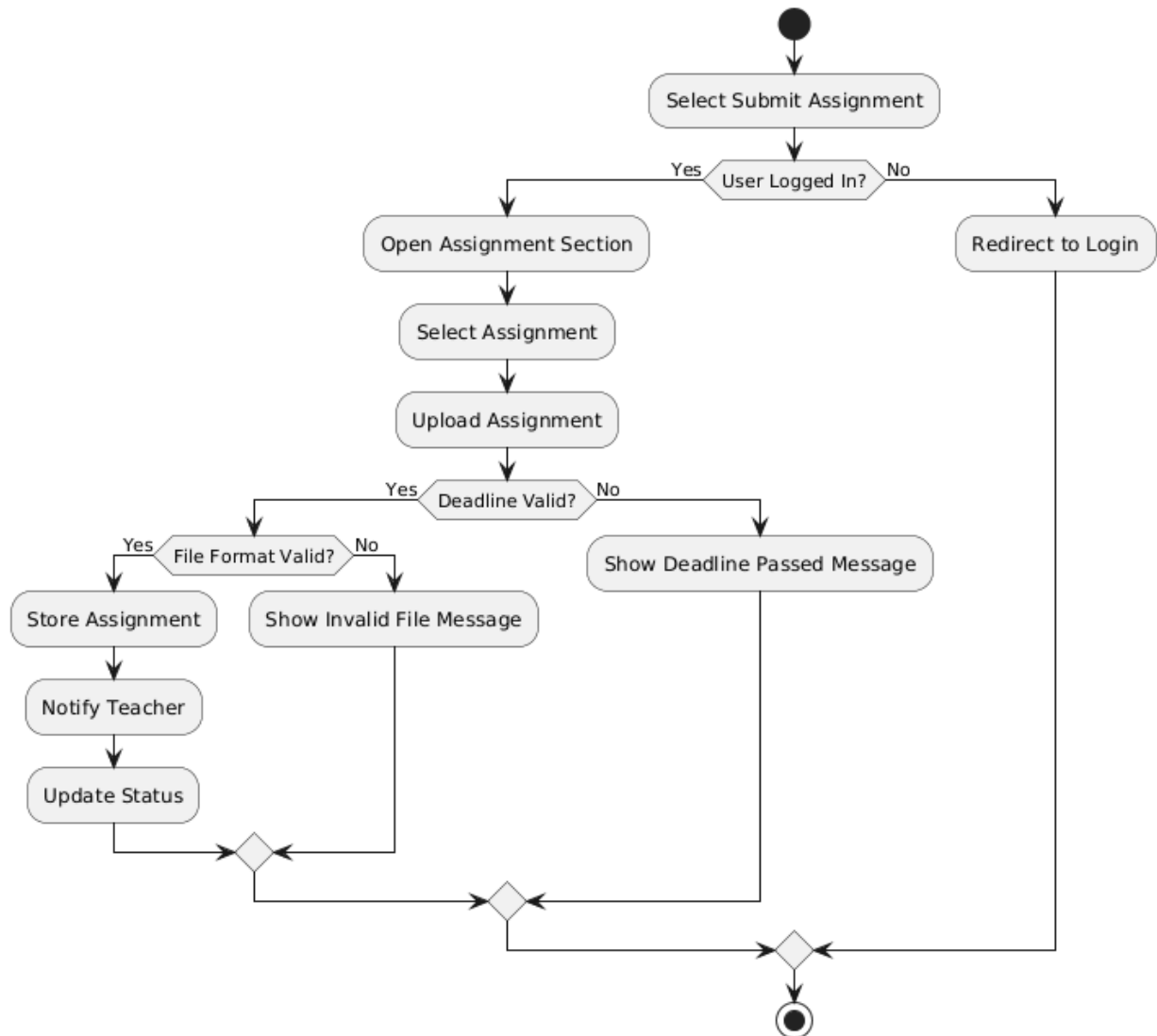
7.5 Activity Diagram of **Submit Assignment**

Figure 6: Activity Diagram for Submit Assignment

7.6 Activity Diagram of Practice Exercise

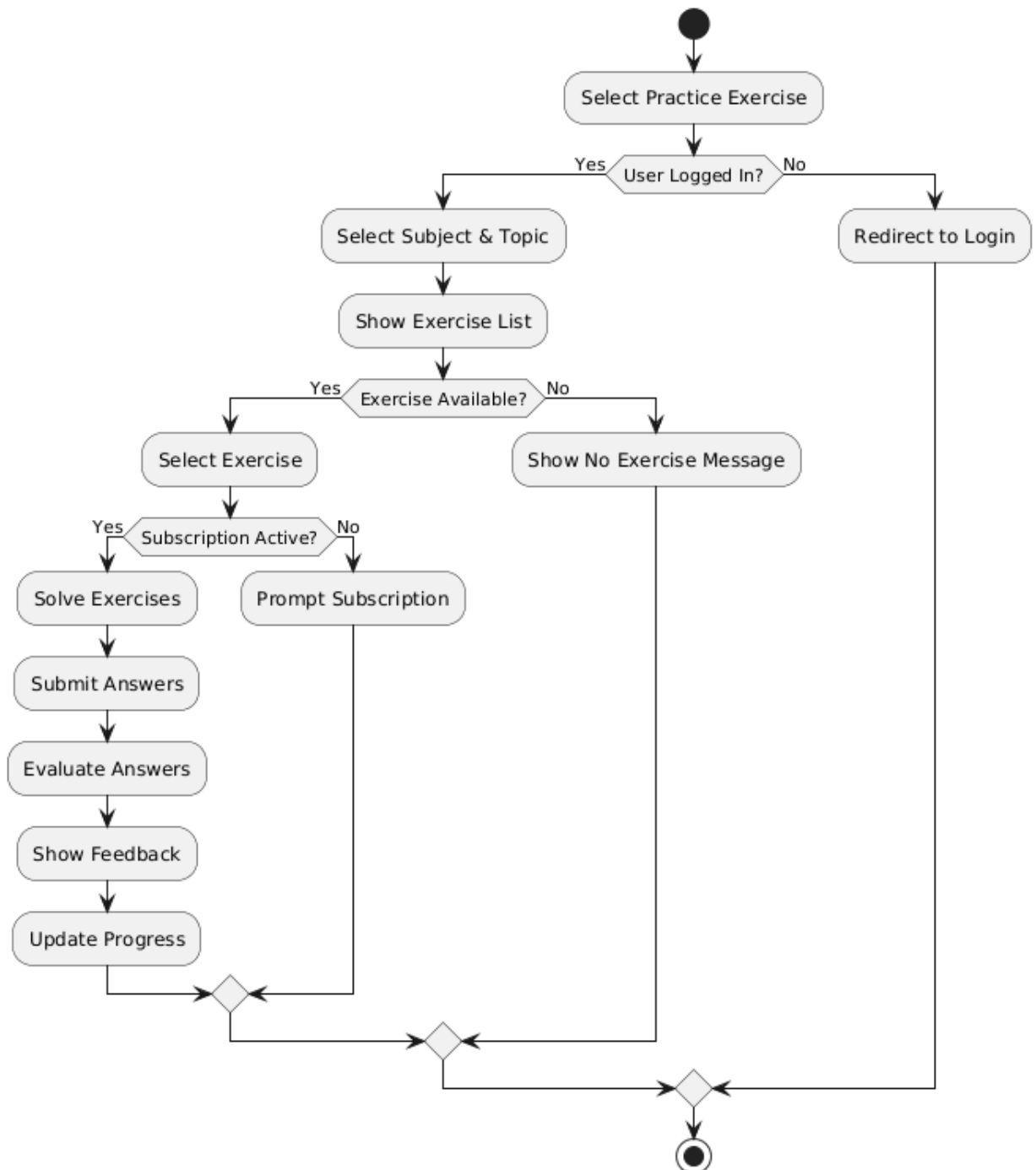


Figure 7: Activity Diagram for Practice Exercise

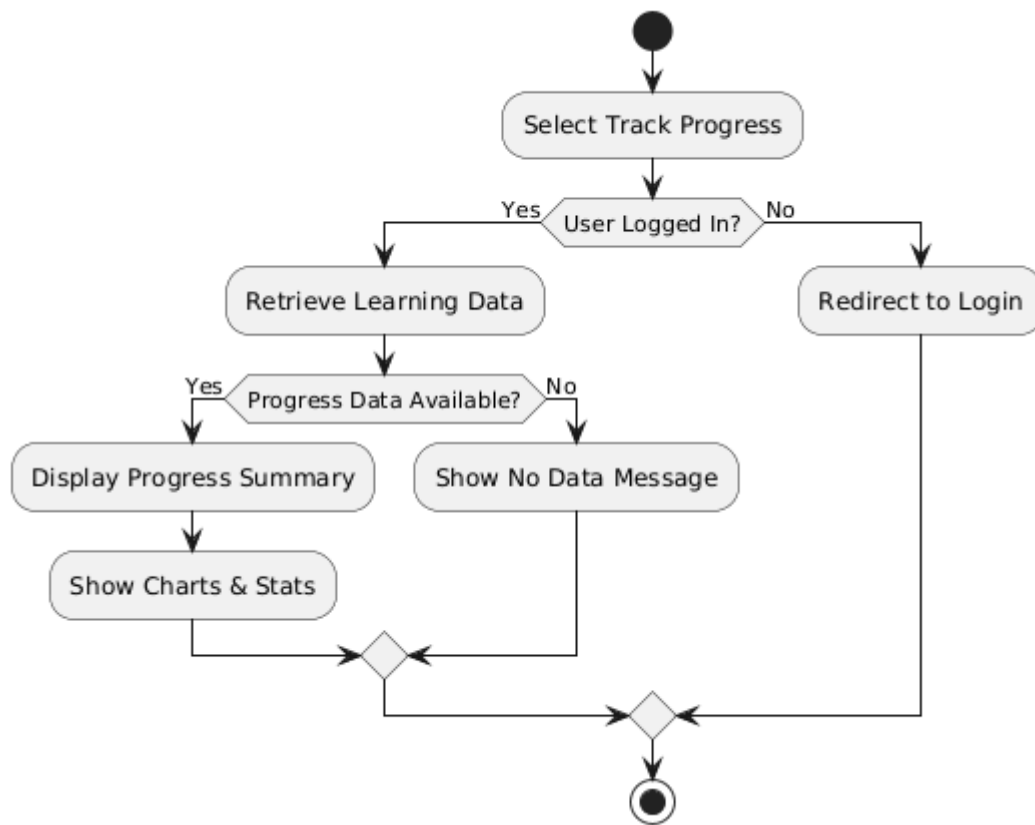
7.7 Activity Diagram of **Track Progress**

Figure 8: Activity Diagram for Track Progress

7.8 Activity Diagram of Manage Subscription

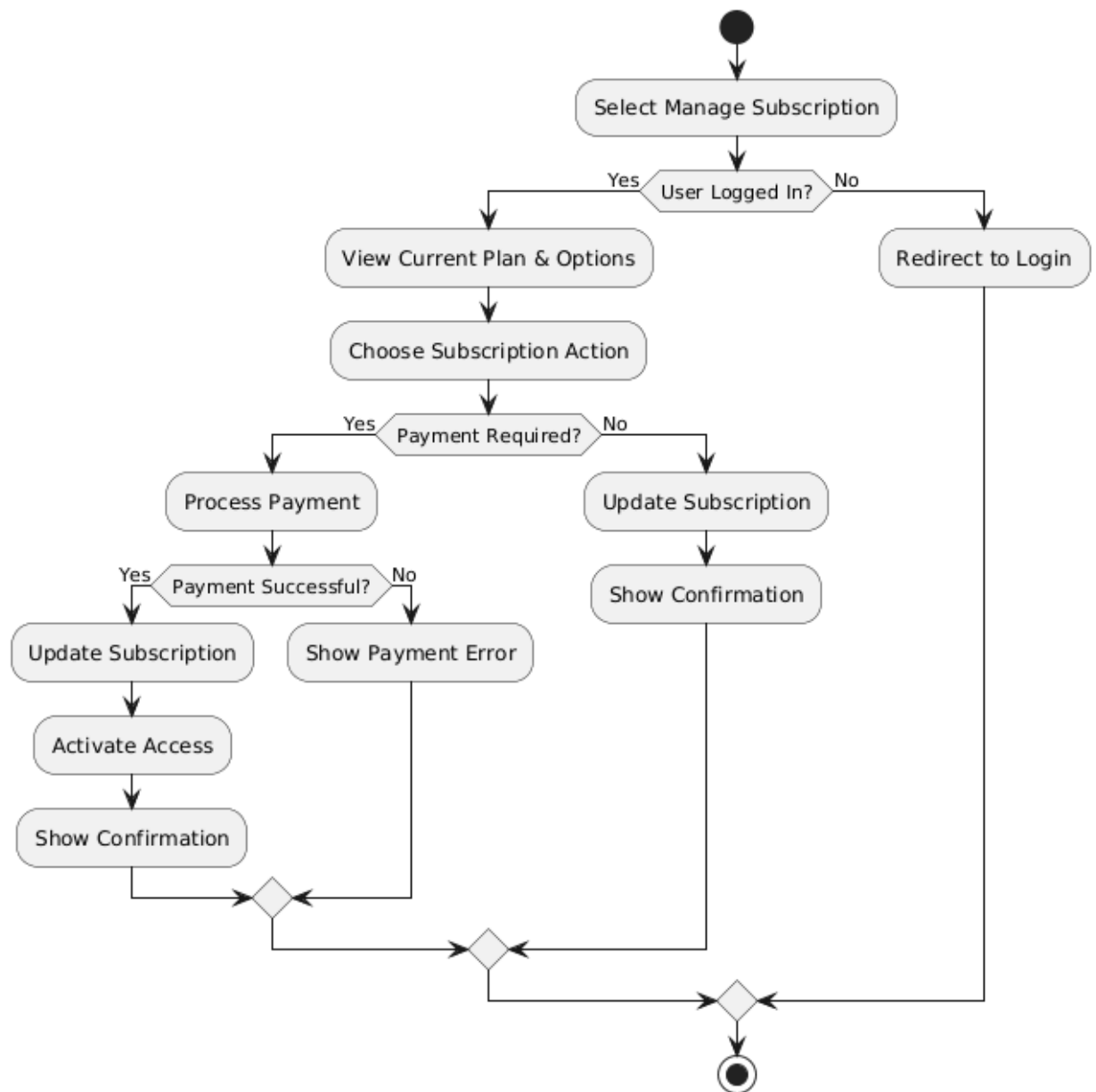


Figure 9: Activity Diagram for Manage Subscription

7.9 Activity Diagram of Upload Content

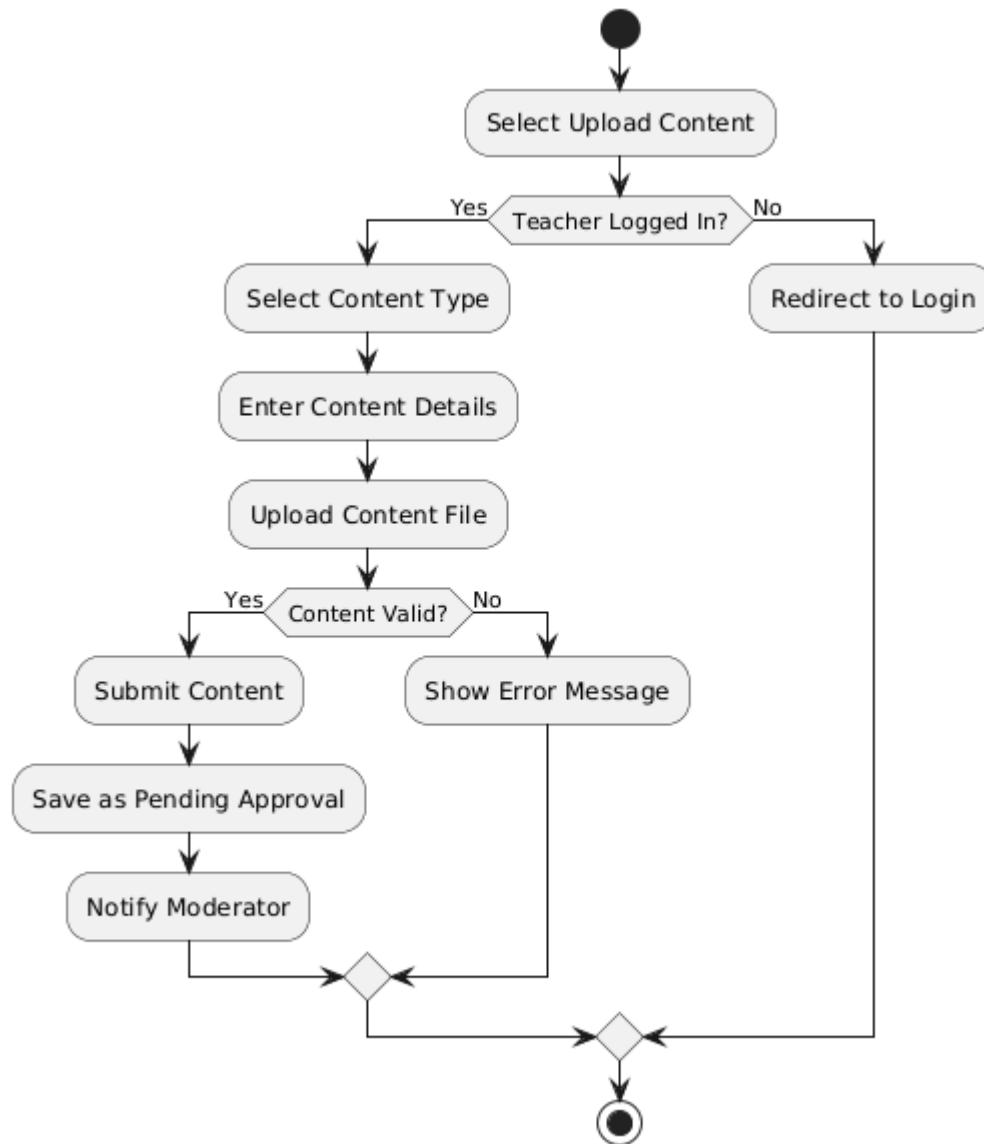
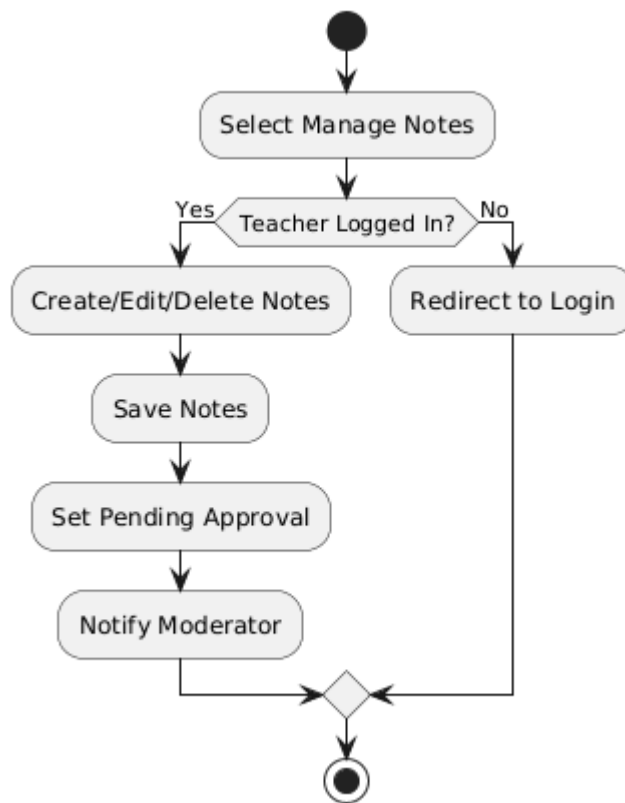
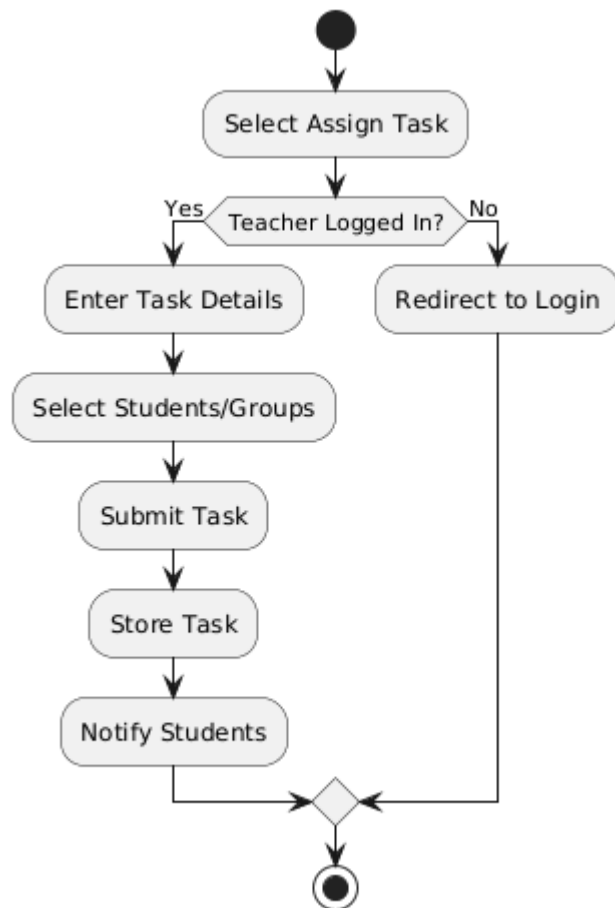
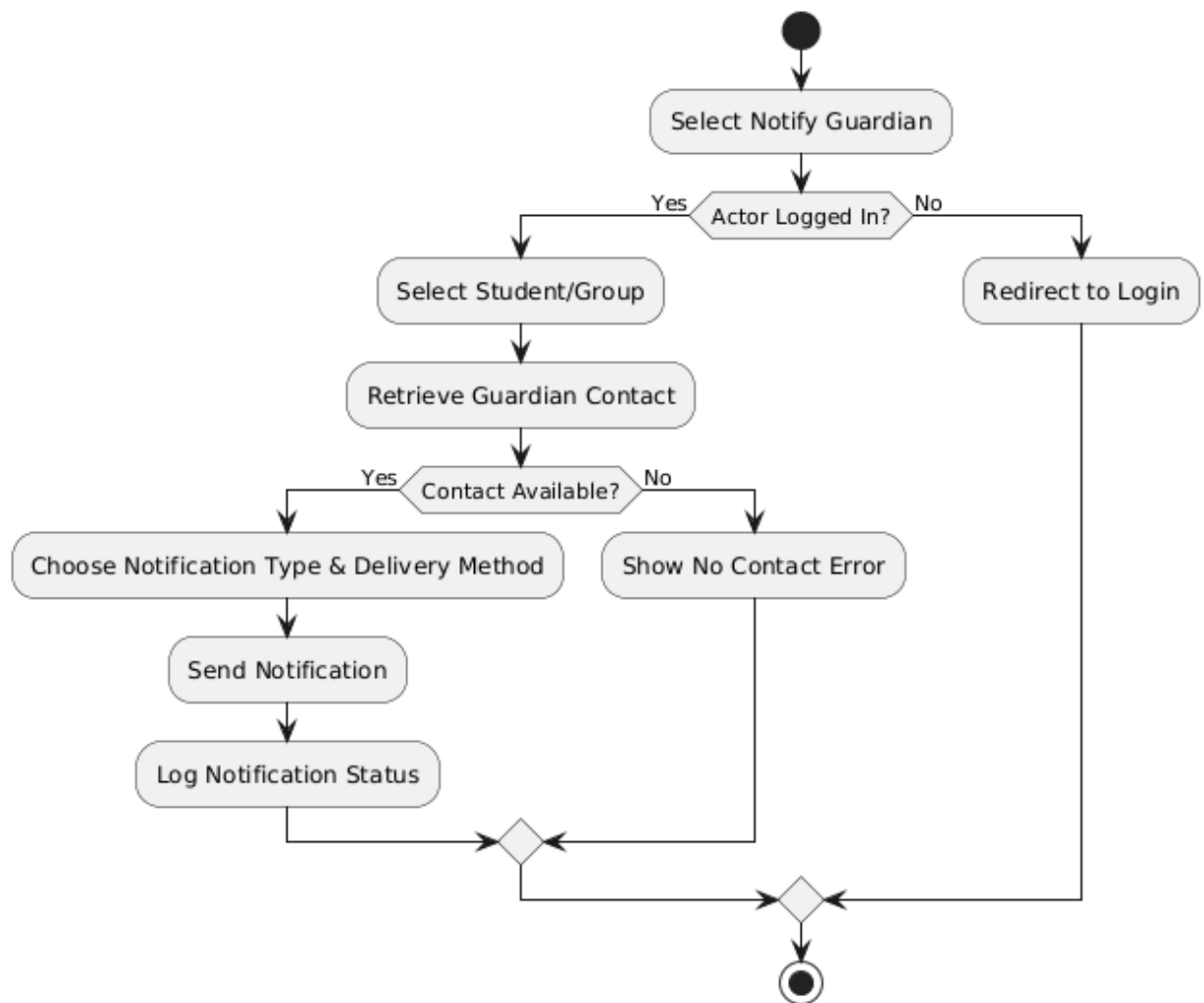
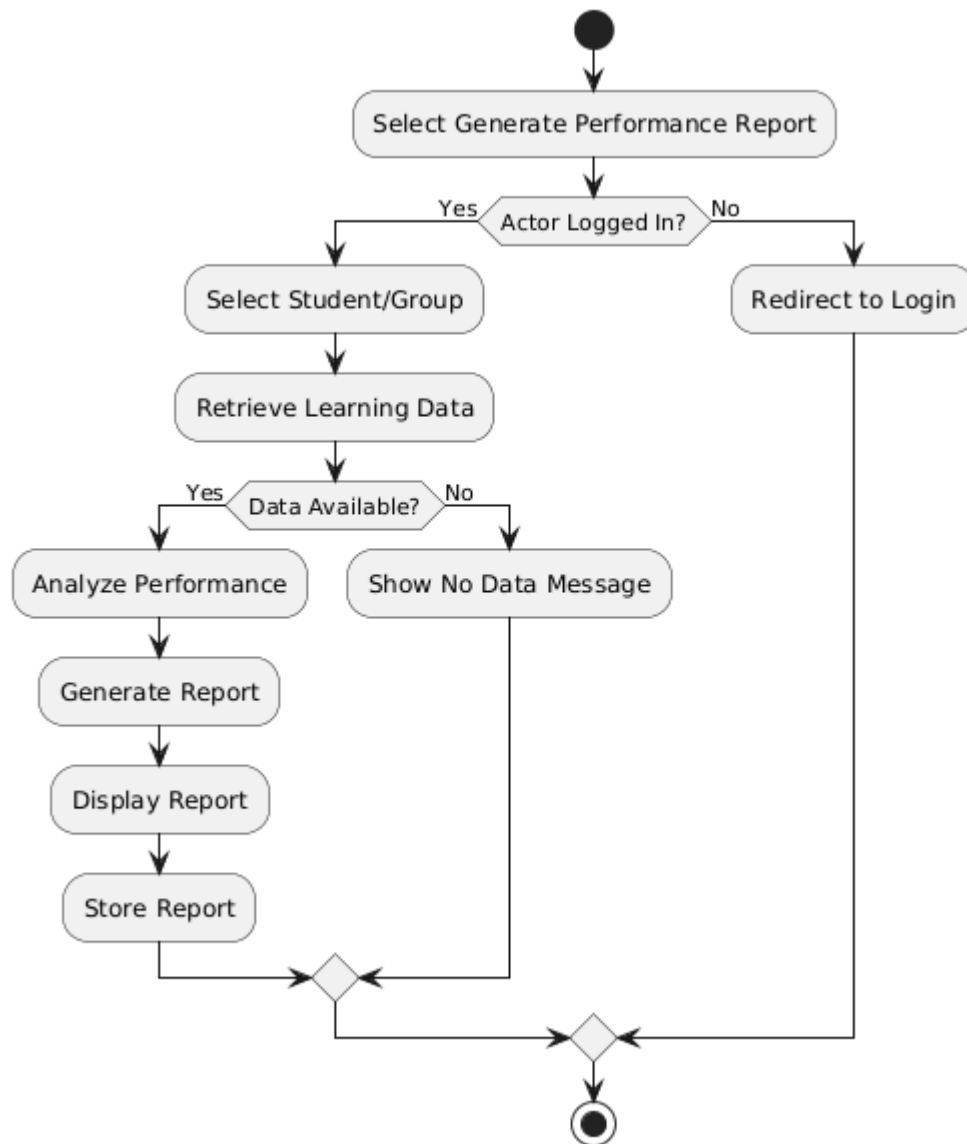


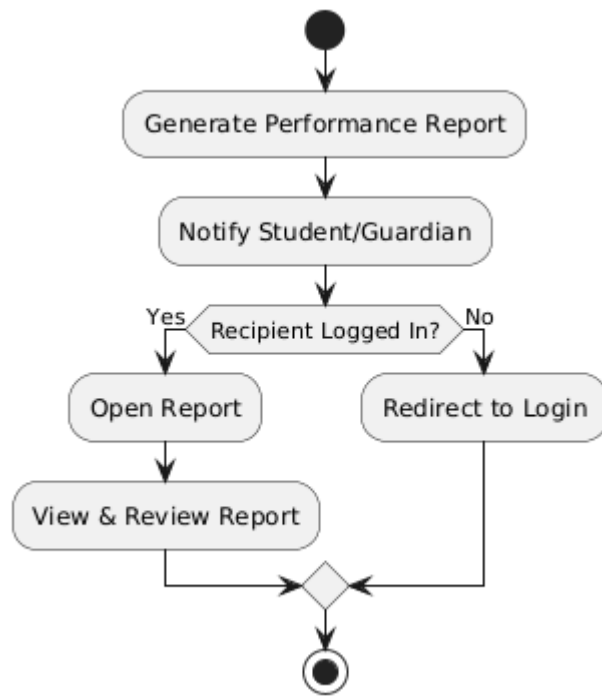
Figure 10: Activity Diagram for Upload Content

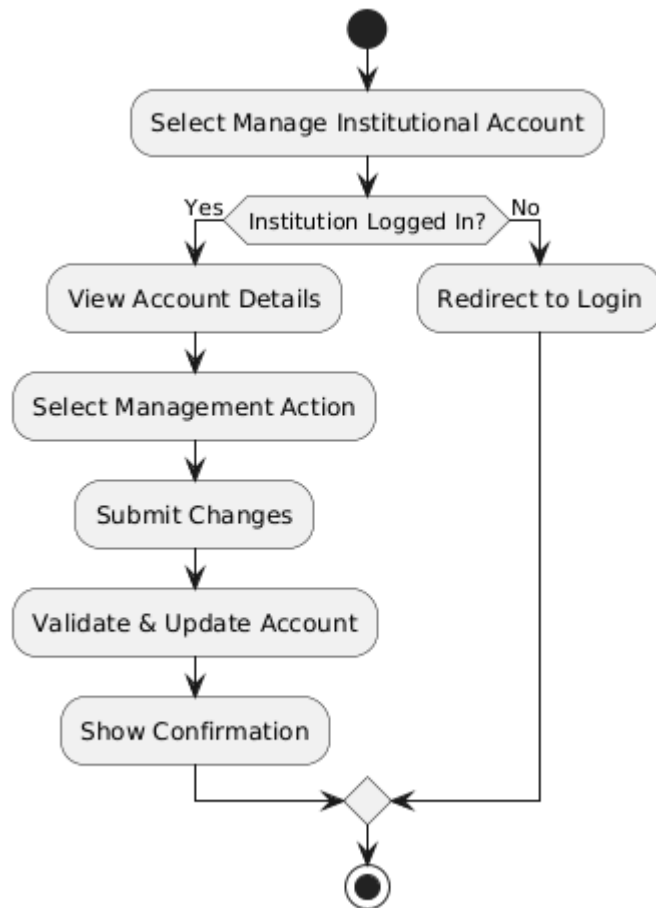
7.10 Activity Diagram of **Manage Notes**Figure 11: Activity Diagram for **Manage Notes**

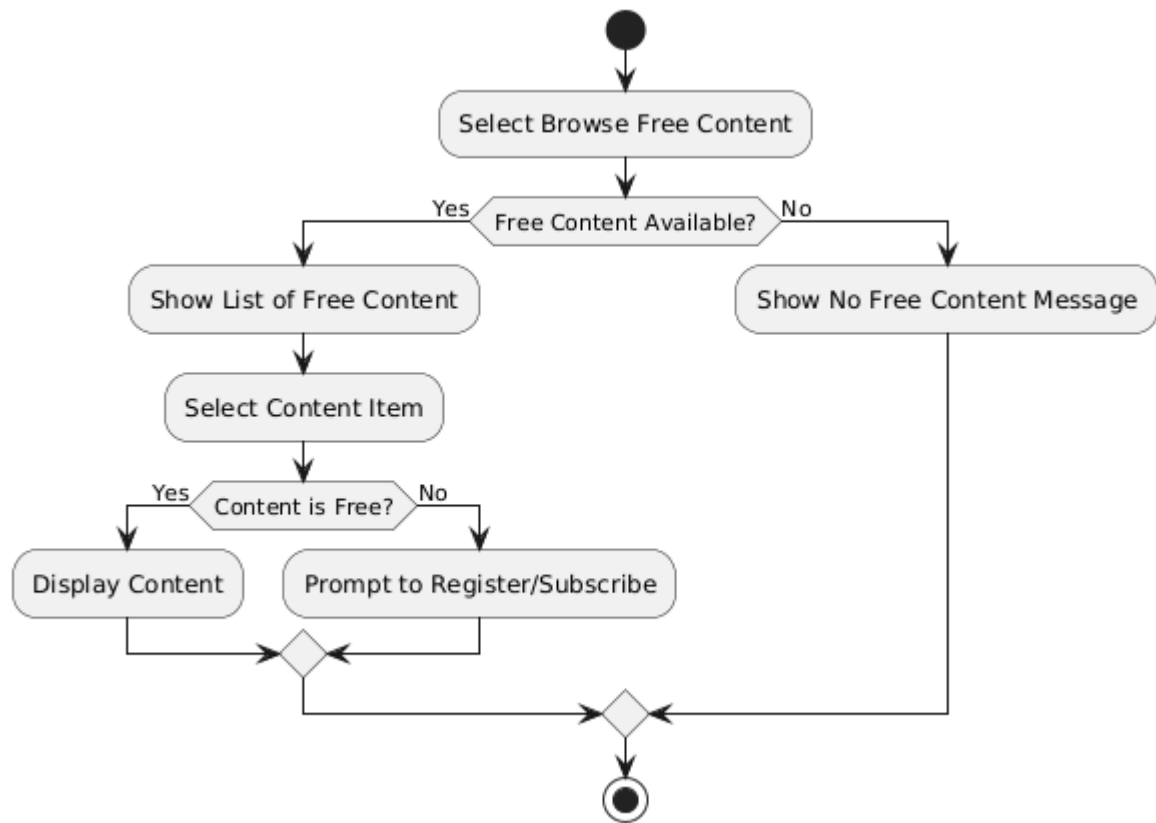
7.11 Activity Diagram of **Assign Task**Figure 12: Activity Diagram for **Assign Task**

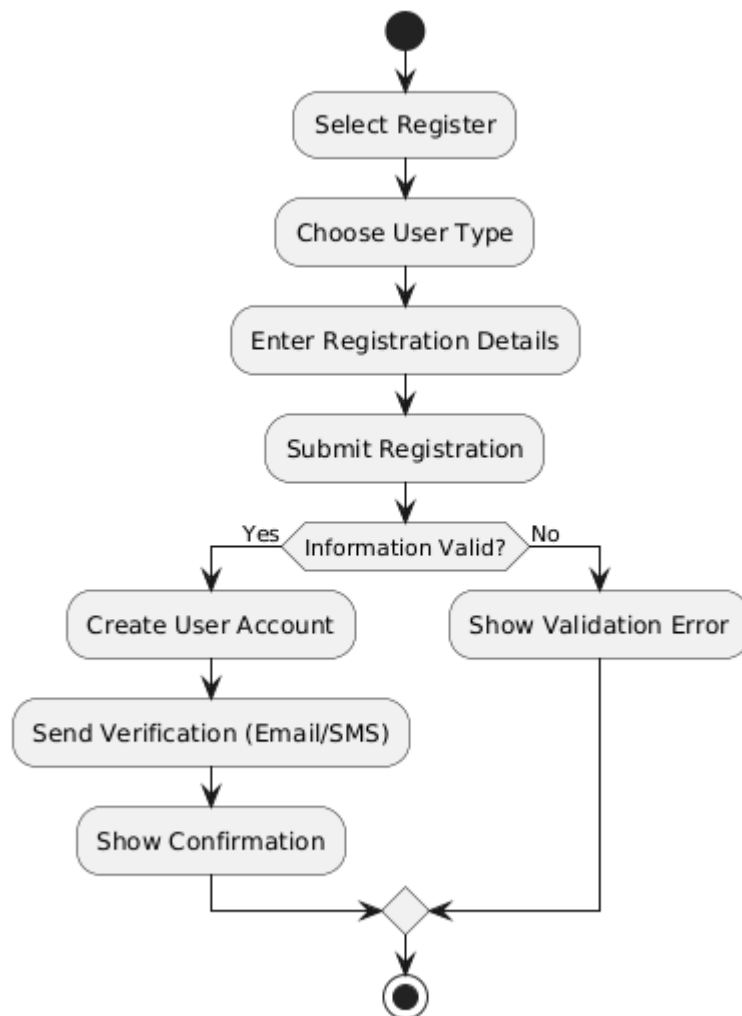
7.12 Activity Diagram of **Notify Guardian**Figure 13: Activity Diagram for **Notify Guardian**

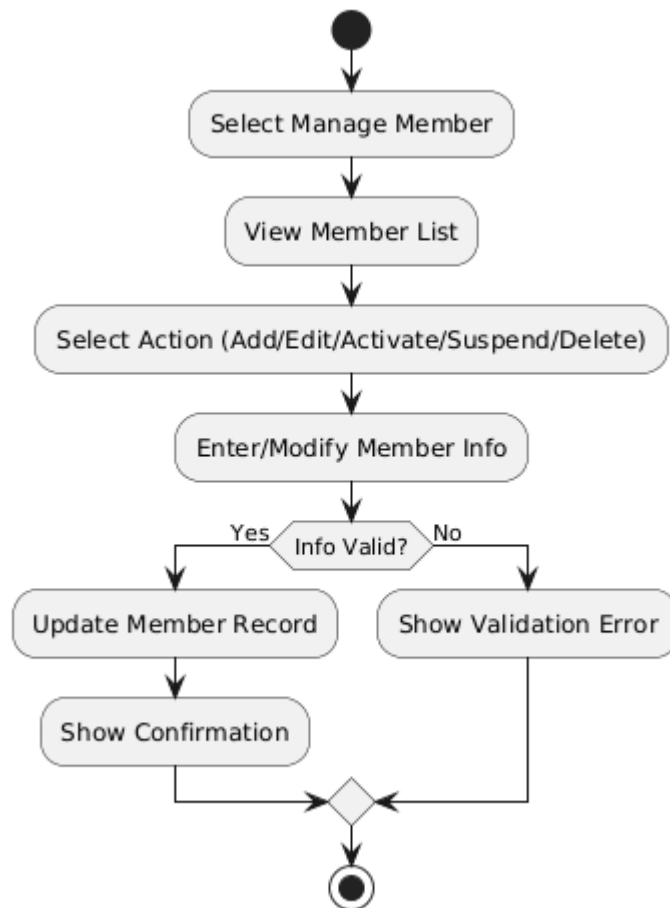
7.13 Activity Diagram of **Generate Performance Report**Figure 14 : Activity Diagram for **Generate Performance Report**

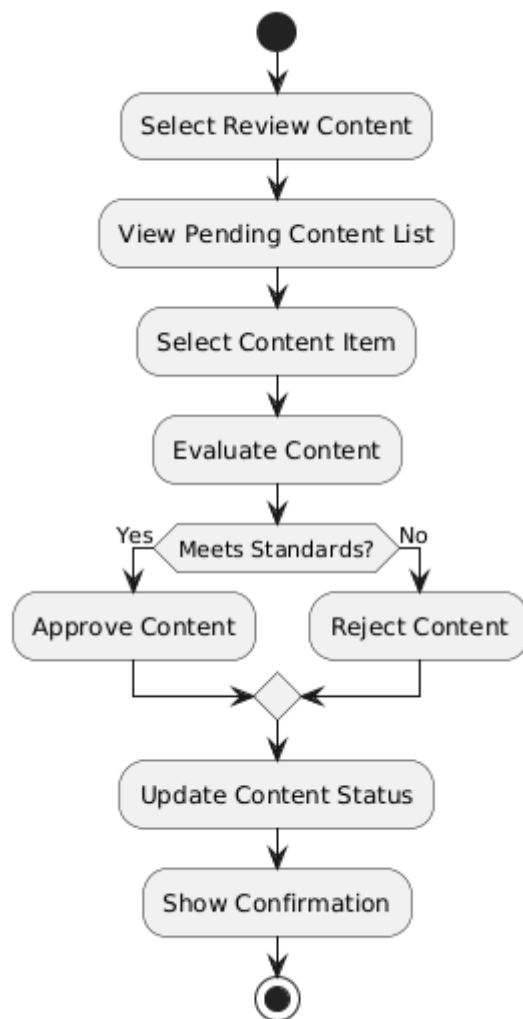
7.14 Activity Diagram for **Receive Performance Report**Figure 15 : Activity Diagram for **Receive Performance Report**

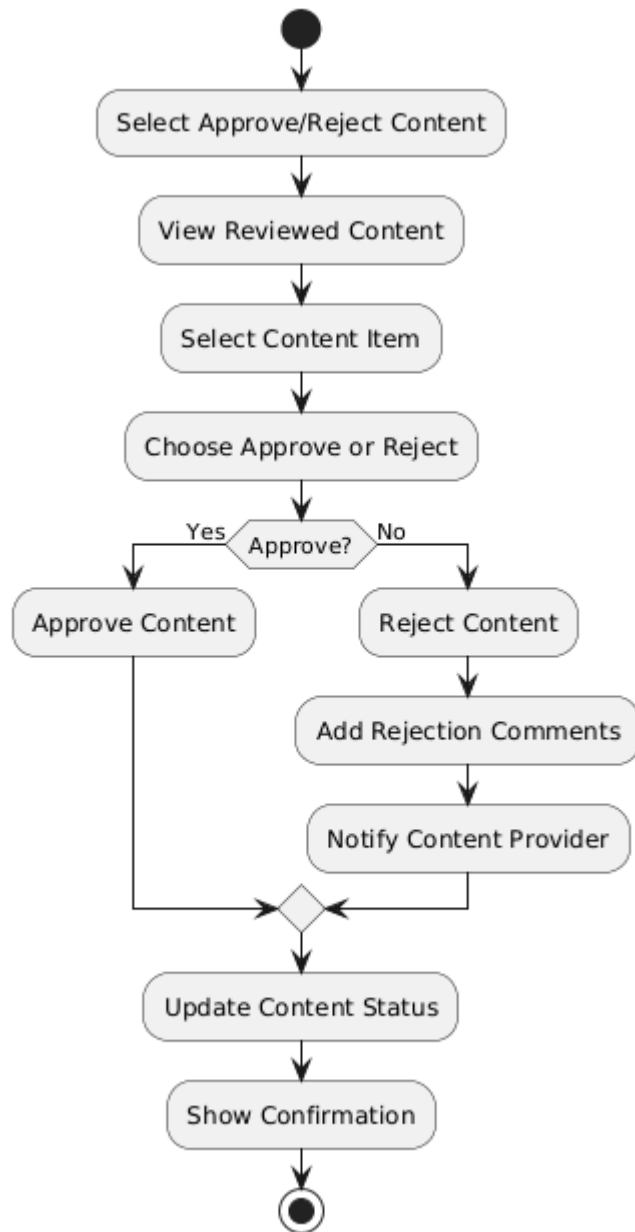
7.15 Activity Diagram of **Manage Institutional Account***Figure 16: Activity Diagram for Manage Institutional Account*

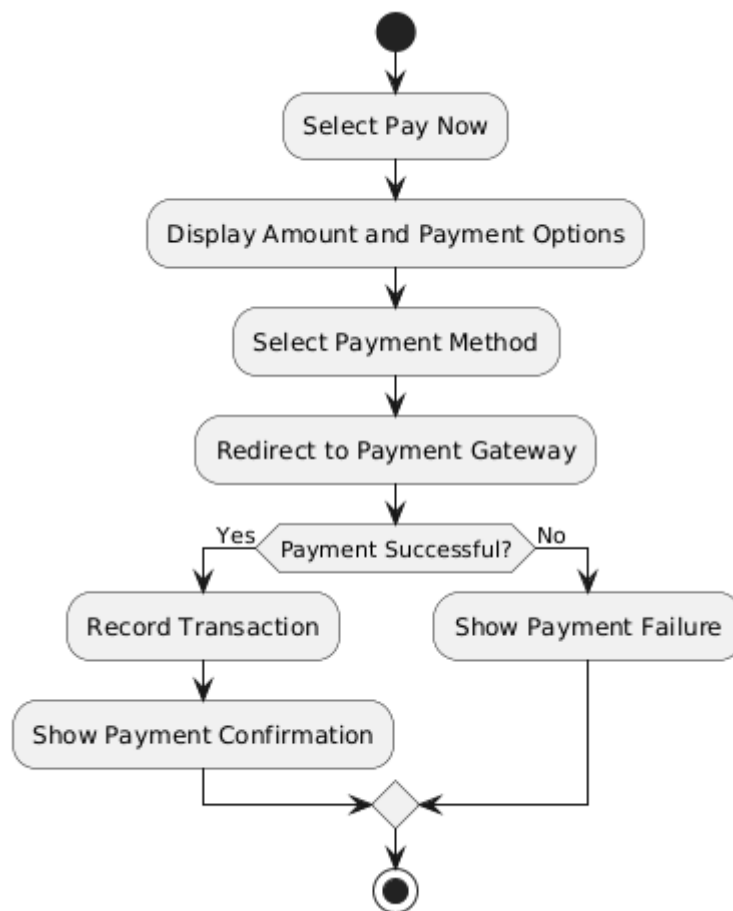
7.16 Activity Diagram of **Browse Free Content (For Guest)**Figure 17: Activity Diagram for **Browse Free Content**

7.17 Activity Diagram of **Register User**Figure 18: Activity Diagram for **Register User**

7.18 Activity Diagram of **Manage Member**Figure 19: Activity Diagram for **Manage Member**

7.19 Activity Diagram of **Review Content**Figure 20: Activity Diagram for **Review Content**

7.20 Activity Diagram of **Approve/Reject Content**Figure 21: Activity Diagram for **Approve/Reject Content**

7.21 Activity Diagram of **Process Payment**Figure 22: Activity Diagram for **Process Payment**

7.22 Activity Diagram of Sent Notification

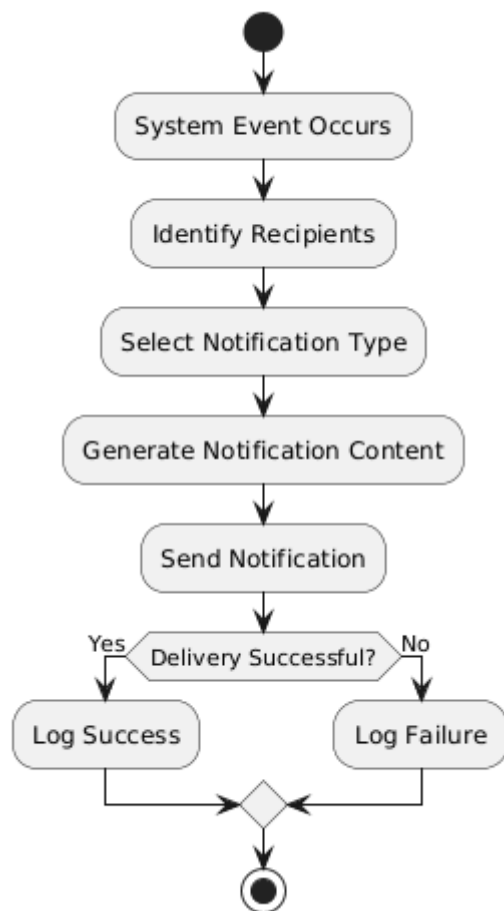


Figure 23: Activity Diagram for Sent Notification

8. DFD

DFD Level 0 - Calcure Context Diagram

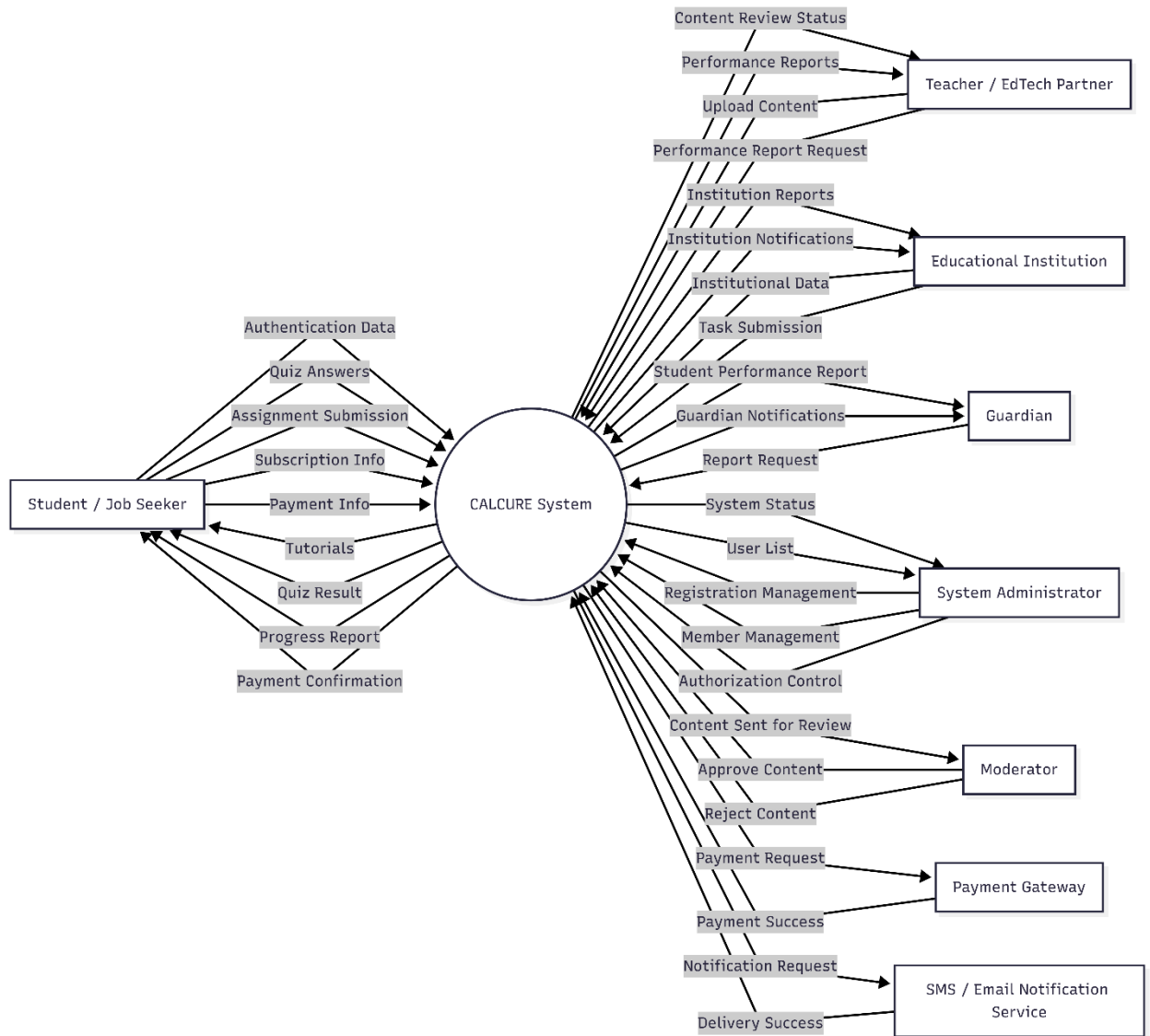


Figure 24: Level 0 DFD

9. RTM (Requirements Traceability Matrix):

Use Case ID's:

- ❖ U1 – Authentication.
- ❖ U2 – Authorization.
- ❖ U3 – View Tutorial.
- ❖ U4 – Attempt Quiz.
- ❖ U5 – Submit Assignment.
- ❖ U6 – Practice Exercises.
- ❖ U7 – Track Progress.
- ❖ U8 – Manage Subscription.
- ❖ U9 – Upload Learning Content.
- ❖ U10 – Manage Notes.
- ❖ U11 – Assign Task.
- ❖ U12 – Generate Performance Report.
- ❖ U13 – Notify Guardian.
- ❖ U14 – Manage Institutional Account.
- ❖ U15 – Receive Performance Report.
- ❖ U16 – Browse Free Content.
- ❖ U17 – Register User.
- ❖ U18 – Manage Member.
- ❖ U19 – Review Content.
- ❖ U20 – Approve / Reject Content.
- ❖ U21 – Process Payment.
- ❖ U22 – Send Notification.

Requirements ID's:

Business Requirements

- ❖ R1 — Subscription-based math learning services.
- ❖ R2 — Verified content marketplace for teachers and partners.

Business Objectives

- ❖ R3 — Improve math learning outcomes through guided practice.
- ❖ R4 — Expand reach via remote delivery and partnerships.

User Requirements**Primary Users**

- ❖ R5 — Student subscription access to premium resources.
- ❖ R6 — Topic-wise learning navigation
- ❖ R7 — Step-by-step solution consumption.
- ❖ R8 — Video tutorial learning.
- ❖ R9 — Assessment and practice participation.
- ❖ R10 — Teacher content and assessment creation.

Secondary Users

- ❖ R11 — Guest access to free features.
- ❖ R12 — Guardian performance reporting.
- ❖ R13 — Institutional subscription and student access control.
- ❖ R14 — Job seeker preparation support.

Business Rules

- ❖ **R15** — Moderator approval required for content publishing.
- ❖ **R16** — Premium access enforcement through valid entitlement.

Functional Requirements

- ❖ **R17** — Account creation and role-based authorization.
- ❖ **R18** — Admin membership management.
- ❖ **R19** — Teacher lesson and content upload.
- ❖ **R20** — Moderation workflow management.
- ❖ **R21** — Content catalog browsing and search.
- ❖ **R22** — Video delivery and playback support.
- ❖ **R23** — Step-by-step solution rendering.
- ❖ **R24** — Quiz management and attempt processing.
- ❖ **R25** — Assignment publishing and submission handling.
- ❖ **R26** — Progress tracking and performance reporting.

Quality Attributes / Performance Requirements

- ❖ **R27** — System availability.
- ❖ **R28** — Acceptable response time.
- ❖ **R29** — Low-bandwidth video streaming support.
- ❖ **R30** — Security and privacy protection.
- ❖ **R31** — System scalability.

External Interface Requirements

- ❖ **R32** — Payment gateway integration.
- ❖ **R33** — SMS and email service integration.

Design Constraints

- ❖ **R34** — Mobile-first user experience.
- ❖ **R35** — Bilingual (Bangla & English) support.
- ❖ **R36** — Standard mathematical notation rendering.
- ❖ **R37** — Content versioning and traceability.
- ❖ **R38** — Protection of minors’ data.

Data Requirements

- ❖ **R39** — User and role data storage
- ❖ **R40** — Content metadata and storage references
- ❖ **R41** — Assessment and submission records
- ❖ **R42** — Subscription and payment records
- ❖ **R43** — Notification and delivery history

[illegible]

[illegible]